16 Thermodynamic potentials

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The internal energy U of a system is a function of state, which means that a system undergoes the same change in U when we move it from one equilibrium state to another, irrespective of which route we take through parameter space. This makes U a very useful quantity, though not a uniquely useful quantity. In fact, we can make a number of other functions of state, simply by adding to U various other combinations of the functions of state p, V, T and S in such a way as to give the resulting quantity the dimensions of energy. These new functions of state are called **thermodynamic potentials**, and examples include $U+TS$, $U-pV$, $U+2pV-3TS$. However, most thermodynamic potentials that one could pick are really not very useful (including the ones we've just used as examples!) but three of them are extremely useful and are given special symbols: $H = U + pV$, $F = U - TS$ and $G = U + pV - TS$. In this chapter, we will explore why these three quantities are so useful. First, however, we will review some properties concerning the internal energy U.

16.1 Internal energy, U

Let us review the results concerning the internal energy that were derived in Section 14.3. Changes in the internal energy U of a system are given by the first law of thermodynamics written in the form (eqn 14.17):

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V. \tag{16.1}$$

[1] See Section 14.3.

This equation shows that the *natural variables*[1] to describe U are S and V, since changes in U are due to changes in S and/or V. Hence we write $U = U(S, V)$ to show that U is a function of S and V. Moreover, if S and V are held constant for the system, then

$$\mathrm{d}U = 0, \tag{16.2}$$

which is the same as saying that U is a constant. Equation 16.1 implies that the temperature T can be expressed as a differential of U using

$$T = \left(\frac{\partial U}{\partial S}\right)_V, \tag{16.3}$$

and similarly the pressure p can be expressed as

$$p = -\left(\frac{\partial U}{\partial V}\right)_S. \tag{16.4}$$

We also have that for **isochoric** processes (where isochoric means that V is constant),

$$\mathrm{d}U = T\mathrm{d}S, \tag{16.5}$$

and for reversible[2] isochoric processes

$$\mathrm{d}U = đQ_{\mathrm{rev}} = C_V\,\mathrm{d}T, \tag{16.6}$$

and hence

$$\Delta U = \int_{T_1}^{T_2} C_V\,\mathrm{d}T. \tag{16.7}$$

This is only true for systems held at constant volume; we would like to be able to extend this to systems held at constant pressure (an easier constraint to apply experimentally), and this can be achieved using the thermodynamic potential called enthalpy which we describe next.

[2] For a reversible process, $đQ = T\,\mathrm{d}S$, see Section 14.3.

16.2 Enthalpy, H

We define the **enthalpy** H by

$$\boxed{H = U + PV.} \tag{16.8}$$

This definition together with eqn 16.1 imply that

$$\begin{aligned} \mathrm{d}H &= T\mathrm{d}S - p\mathrm{d}V + p\mathrm{d}V + V\mathrm{d}p \\ &= T\mathrm{d}S + V\mathrm{d}p. \end{aligned} \tag{16.9}$$

The natural variables for H are thus S and p, and we have that $H = H(S,p)$. We can therefore immediately write down that for a **isobaric** (i.e. constant pressure) process,

$$\mathrm{d}H = T\mathrm{d}S, \tag{16.10}$$

and for a reversible isobaric process

$$\mathrm{d}H = đQ_{\mathrm{rev}} = C_p\,\mathrm{d}T, \tag{16.11}$$

so that

$$\Delta H = \int_{T_1}^{T_2} C_p\,\mathrm{d}T. \tag{16.12}$$

This shows the importance of H, that for reversible isobaric processes the enthalpy represents the heat absorbed by the system.[3] Isobaric conditions are relatively easy to obtain: an experiment which is open to the air in a laboratory is usually at constant pressure since pressure is provided by the atmosphere.[4] We also conclude from eqn 16.9 that if both S and p are constant, we have that $\mathrm{d}H = 0$.

Equation 16.9 also implies that

$$T = \left(\frac{\partial H}{\partial S}\right)_p, \tag{16.13}$$

[3] If you add heat to the system at constant pressure, the enthalpy H of the system goes up. If heat is provided by the system to its surroundings H goes down.

[4] At a given latitude, the atmosphere provides a constant pressure, small changes due to weather fronts notwithstanding.

and

$$V = \left(\frac{\partial H}{\partial p}\right)_S . \tag{16.14}$$

Both U and H suffer from the drawback that one of their natural variables is the entropy S, which is not a very easy parameter to vary in a lab. It would be more convenient if we could substitute that for the temperature T, which is, of course, a much easier quantity to control and to vary. This is accomplished for both of our next two functions of state, the Helmholtz and Gibbs functions.

16.3 Helmholtz function, F

[5]This is sometimes called Helmholtz free energy.

We define the **Helmholtz function**[5] using

$$\boxed{F = U - TS.} \tag{16.15}$$

Hence we find that

$$\begin{aligned} \mathrm{d}F &= T\mathrm{d}S - p\mathrm{d}V - T\mathrm{d}S - S\mathrm{d}T \\ &= -S\mathrm{d}T - p\mathrm{d}V. \end{aligned} \tag{16.16}$$

This implies that the natural variables for F are V and T, and we can therefore write $F = F(T, V)$. For an *isothermal* process (constant T), we can simplify eqn 16.16 further and write that

$$\mathrm{d}F = -p\mathrm{d}V, \tag{16.17}$$

and hence

$$\Delta F = -\int_{V_1}^{V_2} p\mathrm{d}V. \tag{16.18}$$

Hence a positive change in F represents reversible work done on the system by the surroundings, while a negative change in F represents reversible work done on the surroundings by the system. As we shall see in Section 16.5, F represents the maximum amount of work you can get out of a system at constant temperature, since the system will do work on its surroundings until its Helmholtz function reaches a minimum. Equation 16.16 implies that the entropy S can be written as

$$S = -\left(\frac{\partial F}{\partial T}\right)_V , \tag{16.19}$$

and the pressure p as

$$p = -\left(\frac{\partial F}{\partial V}\right)_T . \tag{16.20}$$

If T and V are constant, we have that $\mathrm{d}F = 0$ and F is a constant.

16.4 Gibbs function, G.

We define the **Gibbs function**[6] using

$$\boxed{G = H - TS.} \tag{16.21}$$

[6] This is sometimes called the Gibbs free energy.

Hence we find that

$$\begin{aligned} \mathrm{d}G &= T\mathrm{d}S + V\mathrm{d}p - T\mathrm{d}S - S\mathrm{d}T \\ &= -S\mathrm{d}T + V\mathrm{d}p, \end{aligned} \tag{16.22}$$

and hence the natural variables of G are T and p. [Hence we can write $G = G(T, p)$.]

Having T and p as natural variables is particularly convenient as T and p are the easiest quantities to manipulate and control for most experimental systems. In particular, note that if T and p are constant, $\mathrm{d}G = 0$. Hence G is conserved in any isothermal isobaric process.[7]

[7] For example, at a phase transition between two different phases (call them phase 1 and phase 2), there is phase coexistence between the two phases at the same pressure at the transition temperature. Hence the Gibbs functions for phase 1 and phase 2 must be equal at the phase transition. This will be particularly useful for us in Chapter 28.

The expression in eqn 16.22 allows us to write down expressions for entropy and volume as follows:

$$S = -\left(\frac{\partial G}{\partial T}\right)_p \tag{16.23}$$

and

$$V = -\left(\frac{\partial G}{\partial p}\right)_T. \tag{16.24}$$

We have now defined the four main thermodynamic potentials which are useful in much of thermal physics: the internal energy U, the enthalpy H, the Helmholtz function F and the Gibbs function G. Before proceeding further, we summarize the main equations which we have used so far.

Function of state		Differential	Natural variables	First derivatives	
Internal energy	U	$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$	$U = U(S,V)$	$T = \left(\frac{\partial U}{\partial S}\right)_V$,	$p = -\left(\frac{\partial U}{\partial V}\right)_S$
Enthalpy	$H = U + pV$	$\mathrm{d}H = T\mathrm{d}S + V\mathrm{d}p$	$H = H(S,p)$	$T = \left(\frac{\partial H}{\partial S}\right)_p$,	$V = \left(\frac{\partial H}{\partial p}\right)_S$
Helmholtz function	$F = U - TS$	$\mathrm{d}F = -S\mathrm{d}T - p\mathrm{d}V$	$F = F(T,V)$	$S = -\left(\frac{\partial F}{\partial T}\right)_V$,	$p = -\left(\frac{\partial F}{\partial V}\right)_T$
Gibbs function	$G = H - TS$	$\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p$	$G = G(T,p)$	$S = -\left(\frac{\partial G}{\partial T}\right)_p$,	$V = \left(\frac{\partial G}{\partial p}\right)_T$

Note that to derive these equations quickly, all you need to do is memorize the definitions of H, F and G and the first law in the form $\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$ and the rest can be written down straightforwardly.

Example 16.1

Show that $U = -T^2 \left(\frac{\partial}{\partial T}\right)_V \frac{F}{T}$ and $H = -T^2 \left(\frac{\partial}{\partial T}\right)_p \frac{G}{T}$.

Solution:

Using the expressions

$$S = -\left(\frac{\partial F}{\partial T}\right)_V, \qquad \text{and} \qquad S = -\left(\frac{\partial G}{\partial T}\right)_p,$$

we can write down

$$U = F + TS = F - T\left(\frac{\partial F}{\partial T}\right)_V = -T^2\left(\frac{\partial (F/T)}{\partial T}\right)_V, \tag{16.25}$$

and

$$H = G + TS = G - T\left(\frac{\partial G}{\partial T}\right)_p = -T^2\left(\frac{\partial (G/T)}{\partial T}\right)_p. \tag{16.26}$$

These equations are known as the **Gibbs–Helmholtz equations** and are useful in chemical thermodynamics.

16.5 Availability

We want to try now to work out how to find the equilibrium properties of a system when it is placed in contact with its surroundings. In general, a system is able to exchange heat with its surroundings and also to do work on its surroundings. Let us now consider a system in contact with surroundings which are at temperature T_0 and pressure p_0 (see Fig. 16.1). Let us consider what happens when we transfer energy $\mathrm{d}U$ and volume $\mathrm{d}V$ *from* the surroundings, *to* the system. The internal energy of the *surroundings* changes by $\mathrm{d}U_0$, where

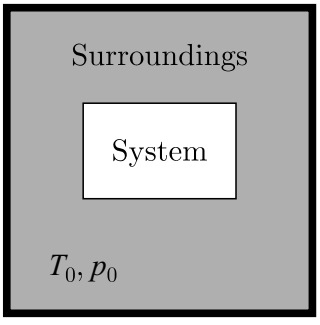


Fig. 16.1 A system in contact with surroundings at temperature T_0 and pressure p_0.

$$\mathrm{d}U_0 = -\mathrm{d}U = T_0\,\mathrm{d}S_0 - p_0(-\mathrm{d}V), \tag{16.27}$$

where the minus signs express the fact that the energy and volume in the surroundings are *decreasing*. We can rearrange this expression to give the change of entropy in the surroundings as

$$\mathrm{d}S_0 = -\left[\frac{\mathrm{d}U + p_0\mathrm{d}V}{T_0}\right]. \tag{16.28}$$

If the entropy of the system changes by $\mathrm{d}S$, then the total change of entropy $\mathrm{d}S_{\mathrm{tot}}$ is

$$\mathrm{d}S_{\mathrm{tot}} = \mathrm{d}S_0 + \mathrm{d}S, \tag{16.29}$$

and the second law of thermodynamics implies that $\mathrm{d}S_{\mathrm{tot}} \geq 0$. Using eqns 16.28 and 16.29, we have that

$$T_0\mathrm{d}S_{\mathrm{tot}} = -\left[\mathrm{d}U + p_0\mathrm{d}V - T_0\mathrm{d}S\right] \geq 0. \tag{16.30}$$

Hence

$$\mathrm{d}U + p_0\mathrm{d}V - T_0\mathrm{d}S \leq 0. \tag{16.31}$$

We now define the **availability** A by

$$A = U + p_0V - T_0S, \tag{16.32}$$

and because p_0 and T_0 are constants, then

$$\mathrm{d}A = \mathrm{d}U + p_0\mathrm{d}V - T_0\mathrm{d}S. \tag{16.33}$$

Hence eqn 16.31 becomes

$$\boxed{\mathrm{d}A \leq 0.} \tag{16.34}$$

We have derived this inequality from the second law of thermodynamics. It demonstrates that changes in A are always negative. As a system settles down to equilibrium, any changes will always force A downwards. Once the system has reached equilibrium, A will be constant at this minimum level. Hence equilibrium can only be achieved by minimizing A. However, the type of equilibrium achieved depends on the nature of the constraints, as we will now show.

- *System with fixed entropy and volume*:
  In this case $\mathrm{d}S = \mathrm{d}V = 0$ and hence eqns 16.33 and 16.34 imply

  $$\mathrm{d}A = \mathrm{d}U \leq 0, \tag{16.35}$$

  so we must *minimize* U to find the equilibrium state of this system.

- *System with fixed entropy and pressure*:
  In this case $\mathrm{d}S = \mathrm{d}p = 0$, and hence eqn 16.33 implies

  $$\mathrm{d}A = \mathrm{d}U + p_0\,\mathrm{d}V. \tag{16.36}$$

  The change in enthalpy is $\mathrm{d}H = \mathrm{d}U + p\,\mathrm{d}V + V\,\mathrm{d}p$, and since $p = p_0$ and $\mathrm{d}p = 0$, we have that

  $$\mathrm{d}H = \mathrm{d}U + p_0\,\mathrm{d}V, \tag{16.37}$$

  and hence

  $$\mathrm{d}A = \mathrm{d}H \leq 0, \tag{16.38}$$

  so we must *minimize* H to find the equilibrium state.

- *System thermally isolated and with fixed volume*:
  Since no heat can enter the system and the system can do no work on its surroundings, $\mathrm{d}U = 0$. Hence eqn 16.33 becomes $\mathrm{d}A = -T_0\,\mathrm{d}S$ and hence $\mathrm{d}A \leq 0$ implies that $\mathrm{d}S \geq 0$. Thus we must *maximise* S to find the equilibrium state.

- *System with fixed volume at constant temperature*:
  $\mathrm{d}A = \mathrm{d}U - T_0\mathrm{d}S \leq 0$, but because $\mathrm{d}T = 0$ and $\mathrm{d}F = \mathrm{d}U - T_0\mathrm{d}S - S\mathrm{d}T = \mathrm{d}U - T_0\mathrm{d}S$, we have that

  $$\mathrm{d}A = \mathrm{d}F \leq 0, \tag{16.39}$$

  so we must *minimize* F to find the equilibrium state.

- *System at constant pressure and temperature*:
  Eqn 16.33 gives $\mathrm{d}A = \mathrm{d}U - T_0\mathrm{d}S + p_0\mathrm{d}V \leq 0$. We can write $\mathrm{d}G$ (from the definition $G = H - TS$) as

$$\mathrm{d}G = \mathrm{d}U + p_0\,\mathrm{d}V + V\,\mathrm{d}p - T_0\,\mathrm{d}S - S\,\mathrm{d}T = \mathrm{d}U - T_0\mathrm{d}S + p_0\mathrm{d}V, \tag{16.40}$$

  since $\mathrm{d}p = \mathrm{d}T = 0$, and hence

$$\mathrm{d}A = \mathrm{d}G \leq 0, \tag{16.41}$$

  so we must *minimize* G to find the equilibrium state.

Example 16.2

Chemistry laboratories are usually at constant pressure. If a chemical reaction is carried out at constant pressure, then by eqn 16.10 we have that

$$\Delta H = \Delta Q, \tag{16.42}$$

and hence ΔH is the reversible heat added to the system, i.e. the heat absorbed by the reaction. (Recall that our convention is that ΔQ is the heat *entering* the system, and in this case the system is the reacting chemicals.)

- If $\Delta H < 0$, the reaction is called **exothermic** and heat will be emitted.
- If $\Delta H > 0$, the reaction is called **endothermic** and heat will be absorbed.

However, this does not tell you whether or not a chemical reaction will actually proceed. Usually reactions occur[8] at constant T and p, so if the system is trying to minimize its availability, then we need to consider ΔG. The second law of thermodynamics (via eqn 16.34 and hence eqn 16.41) therefore implies that a chemical system will minimize G, so that if $\Delta G < 0$, the reaction may spontaneously occur.[9]

[8] The temperature may rise *during* a reaction, but if the final products cool to the original temperature, one only needs to think about the beginning and end points, since G is a function of state.

[9] However, one may also need to consider the kinetics of the reaction. Often a reaction has to pass via a metastable intermediate state which may have a higher Gibbs function, so the system cannot spontaneously lower its Gibbs function without having it slightly raised first. This gives a reaction an **activation energy** which must be added before the reaction can proceed, even though the completion of the reaction gives you all that energy back and more.

16.6 Maxwell's relations

In this section, we are going to derive four equations which are known as Maxwell's relations. These equations are very useful in solving problems in thermodynamics, since each one relates a partial differential between quantities that can be hard to measure to a partial differential between quantities that can be much easier to measure. The derivation proceeds along the following lines: a state functon f is a function of variables x and y. A change in f can be written as

$$\mathrm{d}f = \left(\frac{\partial f}{\partial x}\right)_y \mathrm{d}x + \left(\frac{\partial f}{\partial y}\right)_x \mathrm{d}y. \tag{16.43}$$

Because $\mathrm{d}f$ is an exact differential (see Appendix C.7), we have that

$$\left(\frac{\partial^2 f}{\partial x \partial y}\right) = \left(\frac{\partial^2 f}{\partial y \partial x}\right). \tag{16.44}$$

Hence writing

$$F_x = \left(\frac{\partial f}{\partial x}\right)_y \quad \text{and} \quad F_y = \left(\frac{\partial f}{\partial y}\right)_x, \tag{16.45}$$

we have that

$$\left(\frac{\partial F_y}{\partial x}\right) = \left(\frac{\partial F_x}{\partial y}\right). \tag{16.46}$$

We can now apply this idea to each of the state variables U, H, F and G in turn.

Example 16.3

The Maxwell relation based on G can be derived as follows. We write down an expression for $\mathrm{d}G$:

$$\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p. \tag{16.47}$$

We can also write

$$\mathrm{d}G = \left(\frac{\partial G}{\partial T}\right)_p \mathrm{d}T + \left(\frac{\partial G}{\partial p}\right)_T \mathrm{d}p, \tag{16.48}$$

and hence we can write $S = -\left(\frac{\partial G}{\partial T}\right)_p$ and $V = \left(\frac{\partial G}{\partial p}\right)_T$. Because $\mathrm{d}G$ is an exact differential, we have that

$$\left(\frac{\partial^2 G}{\partial T \partial P}\right) = \left(\frac{\partial^2 G}{\partial p \partial T}\right), \tag{16.49}$$

and hence we have the following Maxwell relation:

$$-\left(\frac{\partial S}{\partial p}\right)_T = \left(\frac{\partial V}{\partial T}\right)_p \tag{16.50}$$

This reasoning can be applied to each of the thermodynamic potentials U, H, F and G to yield the four **Maxwell relations**:

Maxwell's relations:

$$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial p}{\partial S}\right)_V \tag{16.51}$$

$$\left(\frac{\partial T}{\partial p}\right)_S = \left(\frac{\partial V}{\partial S}\right)_p \tag{16.52}$$

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V \tag{16.53}$$

$$\left(\frac{\partial S}{\partial p}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_p. \tag{16.54}$$

These equations should not be memorized;[10] rather it is better to remember how to derive them!

A more sophisticated way of deriving these equations based on Jacobians (which may not to be everybody's taste) is outlined in the box below. It has the attractive virtue of producing all four Maxwell relations in one go by directly relating the work done and heat absorbed in a cyclic process, but the unfortunate vice of requiring easy familiarity with the use of Jacobian transformations.

An alternative derivation of Maxwell's relations:
The following derivation is more elegant, but requires a knowledge of Jacobians (see Appendix C.9): Consider a cyclic process which can be described in both the T–S and p–V planes. The internal energy U is a state function and therefore doesn't change in a cycle, so $\oint p\,\mathrm{d}V = \oint T\,\mathrm{d}S$, and hence we have

$$\int\int \mathrm{d}p\,\mathrm{d}V = \int\int \mathrm{d}T\,\mathrm{d}S, \tag{16.55}$$

so that the work done (the area enclosed by the cycle in the p–V plane) is equal to the heat absorbed (the area enclosed by the cycle in the T–S plane). However, one can also write

$$\int\int \mathrm{d}p\,\mathrm{d}V \frac{\partial(T,S)}{\partial(p,V)} = \int\int \mathrm{d}T\,\mathrm{d}S, \tag{16.56}$$

where $\partial(T,S)/\partial(p,V)$ is the Jacobian of the transformation from the p–V plane to the T–S plane, and so this implies that

$$\boxed{\frac{\partial(T,S)}{\partial(p,V)} = 1.} \tag{16.57}$$

This equation is sufficient to generate all four Maxwell relations via

$$\frac{\partial(T,S)}{\partial(x,y)} = \frac{\partial(p,V)}{\partial(x,y)}, \tag{16.58}$$

where (x,y) are taken as (i) (T,p), (ii) (T,V), (iii) (p,S) and (iv) (S,V), and using the identities in Appendix C.9.

We will now give several examples of how Maxwell's relations can be used to solve problems in thermodynamics.

[10] If you do, however, insist on memorizing them, then lots of mnemonics exist. One useful way of remembering them is as follows. Each Maxwell relation is of the form

$$\left(\frac{\partial *}{\partial \ddagger}\right)_{\star} = \pm\left(\frac{\partial \dagger}{\partial \star}\right)_{\ddagger}$$

where the pairs of symbols which are similar to each other ($\star$ and $*$, or $\dagger$ and $\ddagger$) signify **conjugate variables**, so that their product has the dimensions of energy: e.g. T and S, and p and V. Thus you can notice that, for each Maxwell relation, terms diagonally opposite each other are conjugate variables. The variable held constant is conjugate to the one on the top of the partial differential. Another point is that you always have a minus sign when V and T are on the same side of equation.

Example 16.4

Find expressions for $(\partial C_V/\partial p)_T$ and $(\partial C_V/\partial V)_T$ in terms of p, V and T.
Solution:
By the defintions of C_V and C_p we have that

$$C_V = \left(\frac{\partial Q}{\partial T}\right)_V = T\left(\frac{\partial S}{\partial T}\right)_V \tag{16.59}$$

and

$$C_p = \left(\frac{\partial Q}{\partial T}\right)_p = T\left(\frac{\partial S}{\partial T}\right)_p. \tag{16.60}$$

Now

$$\begin{aligned}\left(\frac{\partial C_p}{\partial p}\right)_T &= \left(\frac{\partial}{\partial p}T\left(\frac{\partial S}{\partial T}\right)_p\right)_T \\ &= T\left(\frac{\partial}{\partial p}\left(\frac{\partial S}{\partial T}\right)_p\right)_T \\ &= T\left(\frac{\partial}{\partial T}\left(\frac{\partial S}{\partial p}\right)_T\right)_p \end{aligned} \tag{16.61}$$

and using a Maxwell relation

$$\left(\frac{\partial C_p}{\partial p}\right)_T = -T\left(\frac{\partial}{\partial T}\left(\frac{\partial V}{\partial T}\right)_p\right)_p = -T\left(\frac{\partial^2 V}{\partial T^2}\right)_p. \tag{16.62}$$

Similarly

$$\left(\frac{\partial C_V}{\partial V}\right)_T = T\left(\frac{\partial^2 p}{\partial T^2}\right)_V. \tag{16.63}$$

Both the expressions in eqns 16.62 and 16.63 are zero for a perfect gas.

Before proceeding further with the examples, we will pause to list the tools which you have at your disposal to solve these sorts of problems. Any given problem may not require you to use all of these, but you may have to use more than one of these 'techniques'.

(1) **Write down a function of state in terms of particular variables**.
If f is a function of x and y, so that $f = f(x, y)$, you then have immediately that

$$\mathrm{d}f = \left(\frac{\partial f}{\partial x}\right)_y \mathrm{d}x + \left(\frac{\partial f}{\partial y}\right)_x \mathrm{d}y. \tag{16.64}$$

(2) **Use Maxwell's relations to transform the partial differential you start with into a more convenient one**.
Use the Maxwell relations in eqns 16.51–16.54.

(3) **Invert a Maxwell relation using the reciprocal theorem**.
The reciprocal theorem states that

$$\left(\frac{\partial x}{\partial z}\right)_y = \frac{1}{\left(\frac{\partial z}{\partial x}\right)_y}, \tag{16.65}$$

and this is proved in Appendix C.6 (see eqn C.41).

(4) **Combine partial differentials using the reciprocity theorem**.
The reciprocity theorem states that

$$\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = -1, \tag{16.66}$$

which is proved in Appendix C.6 (see eqn C.42). This can be combined with the reciprocal theorem to write that

$$\left(\frac{\partial x}{\partial y}\right)_z = -\left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x, \tag{16.67}$$

which is a very useful identity.

(5) **Identify a heat capacity.**
Some of the partial differentials which appear in Maxwell's relations relate to real, measurable properties. As we have seen in Example 16.4, both $\left(\frac{\partial S}{\partial T}\right)_V$ and $\left(\frac{\partial S}{\partial T}\right)_p$ can be related to heat capacities:

$$\frac{C_V}{T} = \left(\frac{\partial S}{\partial T}\right)_V \qquad \text{and} \qquad \frac{C_p}{T} = \left(\frac{\partial S}{\partial T}\right)_p \tag{16.68}$$

(6) **Identify a "generalized susceptibility"**.
A **generalized susceptibility** quantifies how much a particular variable changes when a generalized force is applied. A **generalized force** is a variable such as T or p which is a differential of the internal energy with respect to some other parameter.[11] An example of a generalized susceptibility is $\left(\frac{\partial V}{\partial T}\right)_x$ which, you will recall, answers the question "keeping x constant, how much does the volume change when you change the temperature?". It is related to the thermal expansivity at constant x, where x is pressure or entropy. Thus the **isobaric expansivity** β_p is defined as

$$\beta_p = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p, \tag{16.69}$$

while the **adiabatic expansivity** β_S is defined as

$$\beta_S = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_S. \tag{16.70}$$

Expansivities measure the *fractional* change in volume with a change in temperature.

[11] Recall that $T = (\partial U/\partial S)_V$ and $p = -(\partial U/\partial V)_S$.

Another useful generalized susceptibility is the compressibility. This quantifies how large a fractional volume change you achieve when you apply pressure. The **isothermal compressibility** κ_T is defined as

$$\kappa_T = -\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_T, \tag{16.71}$$

while the **adiabatic compressibility** κ_S is defined as

$$\kappa_S = -\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_S. \tag{16.72}$$

Both quantities have a minus sign so that the compressibilities are positive (this is because things get smaller when you press them, so fractional volume changes are negative when positive pressure is applied). None of these expansivities or compressibilities appears directly in a Maxwell relation, but each can easily be related to those that do using the reciprocal and reciprocity theorems.

Example 16.5

By considering $S = S(T, V)$, show that $C_p - C_V = VT\beta_p^2/\kappa_T$.
Solution:
Considering $S = S(T, V)$ allows us to write down immediately that

$$\mathrm{d}S = \left(\frac{\partial S}{\partial T}\right)_V \mathrm{d}T + \left(\frac{\partial S}{\partial V}\right)_T \mathrm{d}V. \tag{16.73}$$

Differentiating this equation with respect to T at constant p yields

$$\left(\frac{\partial S}{\partial T}\right)_p = \left(\frac{\partial S}{\partial T}\right)_V + \left(\frac{\partial S}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_p. \tag{16.74}$$

Now the first two terms can be replaced by C_p/T and C_V/T respectively, while use of a Maxwell relation and a partial differential identity (see eqn 16.67) yields

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V = -\left(\frac{\partial p}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_p \tag{16.75}$$

and hence using eqns 16.69 and 16.71 we have that

$$C_p - C_V = \frac{VT\beta_p^2}{\kappa_T}. \tag{16.76}$$

The next example shows how to calculate the entropy of an ideal gas.

Example 16.6

Find the entropy of 1 mole of ideal gas.
Solution:
For one mole of ideal gas $pV = RT$. Consider the entropy S as a function of volume and temperature, i.e.

$$S = S(T, V), \tag{16.77}$$

so that

$$\mathrm{d}S = \left(\frac{\partial S}{\partial T}\right)_V \mathrm{d}T + \left(\frac{\partial S}{\partial V}\right)_T \mathrm{d}V \tag{16.78}$$

$$= \frac{C_V}{T}\mathrm{d}T + \left(\frac{\partial p}{\partial T}\right)_V \mathrm{d}V, \tag{16.79}$$

using eqn 16.53 and eqn 16.68. The ideal gas law for 1 mole, $p = RT/V$, implies that

$$\left(\frac{\partial p}{\partial T}\right)_V = R/V, \tag{16.80}$$

and hence, if we integrate eqn 16.79,

$$S = \int \frac{C_V}{T}\mathrm{d}T + \int \frac{R\mathrm{d}V}{V}. \tag{16.81}$$

If C_V is not a function of temperature (which is true for an ideal gas) simple integration yields

$$S = C_V \ln T + R \ln V + \text{constant}. \tag{16.82}$$

The entropy of an ideal gas increases with increasing temperature and increasing volume.

The final example in this chapter shows how to prove that the ratio of the isothermal and adiabatic compressibilities, κ_T/κ_S, is equal to γ.

Example 16.7

Find the ratio of the isothermal and adiabatic compressibilities.
Solution:
This follows using straightforward manipulations of partial differentials. To begin with, we write

$$\frac{\kappa_T}{\kappa_S} = \frac{\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_T}{\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_S}, \tag{16.83}$$

which follows from the definition of κ_T and κ_S (eqns 16.71 and 16.72). Then we proceed as follows:

$$\begin{aligned}
\frac{\kappa_T}{\kappa_S} &= \frac{-\left(\dfrac{\partial V}{\partial T}\right)_p \left(\dfrac{\partial T}{\partial p}\right)_V}{-\left(\dfrac{\partial V}{\partial S}\right)_p \left(\dfrac{\partial S}{\partial p}\right)_V} && \text{reciprocity theorem} \\
&= \frac{\left(\dfrac{\partial V}{\partial T}\right)_p \left(\dfrac{\partial S}{\partial V}\right)_p}{\left(\dfrac{\partial p}{\partial T}\right)_V \left(\dfrac{\partial S}{\partial p}\right)_V} && \text{Maxwell's relations} \\
&= \frac{\left(\dfrac{\partial S}{\partial T}\right)_p}{\left(\dfrac{\partial S}{\partial T}\right)_V} && \text{reciprocity theorem} \\
&= \frac{C_p}{C_V} \\
&= \gamma. && (16.84)
\end{aligned}$$

We can show that this equation is correct for the case of an ideal gas as follows. Assuming the ideal gas equation $pV \propto T$, we have for constant temperature that

$$\frac{\mathrm{d}p}{p} = -\frac{\mathrm{d}V}{V}, \tag{16.85}$$

and hence

$$\kappa_T = \frac{1}{p}. \tag{16.86}$$

For an adiabatic change $p \propto V^{-\gamma}$ and hence

$$\frac{\mathrm{d}p}{p} = -\gamma\frac{\mathrm{d}V}{V}, \tag{16.87}$$

and hence

$$\kappa_S = \frac{1}{\gamma p}. \tag{16.88}$$

This agrees with eqn 16.84 above. We note that because κ_T is larger than κ_S (because $\gamma > 1$), the isotherms always have a smaller gradient than the adiabats on a p–V plot (see Fig. 12.1).

Chapter summary

- We define the following thermodynamic potentials:

$$U, \qquad H = U + pV, \qquad F = U - TS, \qquad G = H - TS,$$

which are then related by the following differentials:

$$\boxed{\begin{aligned} \mathrm{d}U &= T\mathrm{d}S - p\mathrm{d}V \\ \mathrm{d}H &= T\mathrm{d}S + V\mathrm{d}p \\ \mathrm{d}F &= -S\mathrm{d}T - p\mathrm{d}V \\ \mathrm{d}G &= -S\mathrm{d}T + V\mathrm{d}p \end{aligned}}$$

- The availability A is given by $A = U + p_0V - T_0S$, and for any spontaneous change we have that $\mathrm{d}A \leq 0$. This means that a system in contact with a reservoir (temperature T_0, pressure p_0) will minimize A which means
  - minimizing U when S and V are fixed;
  - minimizing H when S and p are fixed;
  - minimizing F when T and V are fixed;
  - minimizing G when T and p are fixed.
- Four Maxwell relations can be derived from the boxed equations above, and used to solve many problems in thermodynamics.

Exercises

(16.1) **(a)** Using the first law $\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$ to provide a reminder, write down the definitions of the four thermodynamic potentials U, H, F, G (in terms of U, S, T, p, V), and give $\mathrm{d}U$, $\mathrm{d}H$, $\mathrm{d}F$, $\mathrm{d}G$ in terms of T, S, p, V and their derivatives.

(b) Derive all the Maxwell relations.

(16.2) **(a)** Derive the following general relations

$$\text{(i)} \quad \left(\frac{\partial T}{\partial V}\right)_U = -\frac{1}{C_V}\left[T\left(\frac{\partial p}{\partial T}\right)_V - p\right]$$

$$\text{(ii)} \quad \left(\frac{\partial T}{\partial V}\right)_S = -\frac{1}{C_V}T\left(\frac{\partial p}{\partial T}\right)_V$$

$$\text{(iii)} \quad \left(\frac{\partial T}{\partial p}\right)_H = \frac{1}{C_p}\left[T\left(\frac{\partial V}{\partial T}\right)_p - V\right]$$

In each case the quantity on the left hand side is the appropriate thing to consider for a particular type of expansion. State what type of expansion each refers to.

(b) Using these relations, verify that for an ideal gas $(\partial T/\partial V)_U = 0$ and $(\partial T/\partial p)_H = 0$, and that $(\partial T/\partial V)_S$ leads to the familiar relation $pV^\gamma =$ constant along an isentrope.

(16.3) Use the first law of thermodynamics to show that

$$\left(\frac{\partial U}{\partial V}\right)_T = \frac{C_p - C_V}{V\beta_p} - p, \tag{16.89}$$

where β_p is the coefficient of volume expansivity and the other symbols have their usual meanings.

(16.4) (a) The natural variables for U are S and V. This

means that if you know S and V, you can find $U(S,V)$. Show that this also gives you simple expressions for T and p.
(b) Suppose instead that you know V, T and the function $U(T,V)$ (i.e. you have expressed U in terms of variables which are not all the natural variables of U). Show that this leads to a (much more complicated) expression for p, namely

$$\frac{p}{T} = \int \left(\frac{\partial U}{\partial V}\right)_T \frac{\mathrm{d}T}{T^2} + f(V), \tag{16.90}$$

where $f(V)$ is some (unknown) function of V.

(16.5) Use thermodynamic arguments to obtain the general result that, for any gas at temperature T, the pressure is given by

$$P = T\left(\frac{\partial P}{\partial T}\right)_V - \left(\frac{\partial U}{\partial V}\right)_T, \tag{16.91}$$

where U is the total energy of the gas.

(16.6) Show that another expression for the entropy per mole of an ideal gas is

$$S = C_p \ln T - R \ln p + \text{constant}. \tag{16.92}$$

(16.7) Show that the entropy of an ideal gas can be expressed as

$$S = C_V \ln\left(\frac{p}{\rho^\gamma}\right) + \text{constant}. \tag{16.93}$$

Hermann von Helmholtz (1821–1894)

Since his family couldn't afford to give him an academic education in physics, the seventeen-year old Helmholtz found himself at a Berlin medical school getting a free four-year medical education, the catch being that he then had to serve as a surgeon in the Prussian army. It was during his time in the army that he submitted a scientific paper 'On the conservation of force' (his use of the word 'force' is more akin to what we call 'energy', the two concepts being poorly differentiated at the time). It was a blow against the notion of a 'vital force', an indwelling 'life source' which was widely proposed by physiologists to explain biological systems. Helmholtz intuited that such a vital force was mere metaphysical speculation and instead all physical and chemical processes involved the exchange of energy from one form to another, and that 'all organic processes are reducible to physics'. Thus he began a remarkable career based on his remarkable physical insight into physiology.



Fig. 16.2 H. von Helmholtz

In 1849 he was appointed professor of physiology at Königsberg, and six years later took up a professorship in anatomy in Bonn, moving to Heidelberg three years later. During this period he pioneered the application of physical and mathematical techniques to physiology: he invented the opthalmoscope (for looking into the eye), the opthalmometer (for measuring the curvature and refractive errors in the eye) and worked on the problem of three-colour vision; he did pioneering research in physiological acoustics, explaining the operation of the inner ear; he also measured the speed of nerve impulses in a frog.

He even found time to make important contributions in understanding vortices in fluids. In 1871, he was appointed to a chair in Berlin, but this time it was in physics; here he pursued work in electrodynamics, non-Euclidean geometry and physical chemistry. Helmholtz mentored and influenced many highly talented students in Berlin, including Planck, Wien and Hertz.

Helmholtz's scientific life was characterized by a search for unity and clarity. He once said that 'whoever in the pursuit of science, seeks after immediate practical utility may rest assured that he seeks in vain', but there can be only few scientists in history whose work has had the result of greater practical utility.

William Thomson [Lord Kelvin] (1824–1907)

William Thomson was something of a prodigy: born in Belfast, the son of a mathematician, he studied in Glasgow University and then moved to Peterhouse, Cambridge. By the time he had graduated, he had written 12 research papers, the first of the 661 of his career. He became Professor of Natural Philosophy in the University of Glasgow at 22, a Fellow of the Royal Society at 27, was knighted at 42, and in 1892 became Baron Kelvin of Largs (taking his new title from the River Kelvin in Glasgow), an appointment which occurred during his presidency of the Royal Society. When he died, he was buried next to Isaac Newton in Westminster Abbey.

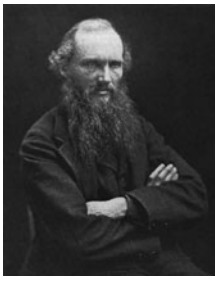


Fig. 16.3 William Thomson

Thomson made pioneering contributions in fundamental electromagnetism and fluid dynamics, but also involved himself in large engineering projects. After working out how to solve the problem of sending signals down very long cables, he was involved in laying the first transatlantic telegraph cables in 1866. In 1893, he headed an international commission to plan the design of the Niagara Falls power station and was convinced by Nikola Tesla, somewhat against his better judgement, to use three-phase AC power rather than his preferred DC power transmission. On this point he was unable to accurately forecast the future (which was of course AC, not DC); in similar vein, he pronounced heavier-than-air flying machines 'impossible', thought that radio had 'no future' and

that war, as a 'relic of barbarism' would 'become as obsolete as duelling'. If only.

It is his progress in thermodynamics that interests us here. Inspired by the meticulous thermometric measurements of Henri Regnault which he had observed during a postgraduate stay in Paris, Thomson proposed an absolute temperature scale in 1848. Thomson was also profoundly influenced by Fourier's theory of heat (which he had read in his teens) and Carnot's work via the paper of Clapeyron. These had assumed a caloric theory of heat, which Thomson had initially adopted, but his encounter with Joule at the 1847 British Association meeting in Oxford had sown some seeds of doubt in caloric. After much thought, Thomson groped his way towards his 'dynamical theory of heat' which he published in 1851, a synthesis of Joule and Carnot, containing a description of the degradation of energy and speculations about the heat death of the Universe. He just missed a full articulation of the concept of entropy, but grasped the essential details of the first and second laws of thermodynamics. His subsequent fruitful collaboration with Joule led to the Joule-Thomson (or Joule-Kelvin) effect.

Thomson also discovered many key results concerning thermoelectricity. His most controversial result was however his estimate of the age of the Earth, based on Fourier's thermal diffusion equation. He concluded that if the Earth had originally been a red-hot globe, and had cooled to his present temperature, its age must be about 10^8 years. This pleased nobody: the Earth was too old for those who believed in a six-thousand year old planet but too young for Darwin's evolution to produce the present biological diversity. Thomson could not have known that radioactivity (undiscovered until the very end of the nineteenth century) acts as an additional heat source in the Earth, allowing the Earth to be nearly two orders of magnitude older than he estimated. His lasting legacy however has been his new temperature scale, so that his 'absolute zero', the lowest possible temperature obtainable, is zero degrees *Kelvin*.

Josiah Willard Gibbs (1839–1903)

Willard Gibbs was born in New Haven and died in New Haven, living his entire life (a brief postdoctoral period in France and Germany excepted) at Yale, where he remained unmarried. His father was also called Josiah Willard Gibbs and had also been a professor at Yale, though in Sacred Literature rather than in mathematical physics. Willard Gibbs' life was quiet and secluded, well away from the centres of intense scientific activity at the time, which were all in Europe. This gave this gentle and scholarly man the opportunity to perform clear-thinking, profound and independent work in chemical thermodynamics, work which turned out to be completely revolutionary, though this took time to be appreciated. Willard Gibbs' key papers were published in a series of installments in the *Transactions of the Conneticut Academy of Sciences*, which was hardly required reading at the time; moreover his mathematical style did not make his papers easily accessible. Maxwell was one of the few who were very impressed.

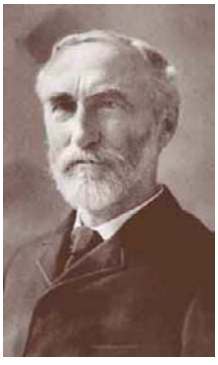


Fig. 16.4 J. W. Gibbs

Gibbs established the key principles of chemical thermodynamics, defined the free energy and chemical potential, completely described phase equilibria with more than one component and championed a geometric view of thermodynamics. Not only did he substantially formulate thermodynamics and statistical mechanics in the form we know it today, but he also championed the use of vector calculus, in its modern form, to describe electromagnetism (in the face of spirited opposition from various prominent Europeans who maintained that the only way to describe electromagnetism was using quaternions).

Gibbs didn't interact a great deal with scientific colleagues in other institutions; he was privately secure in himself and in his ideas. One contemporary wrote of him: 'Unassuming in manner, genial and kindly in his intercourse with his fellow-men, never showing impatience or irritation, devoid of personal ambition of the baser sort or of the slightest desire to exalt himself, he went far toward realising the ideal of the unselfish, Christian gentleman'.

The third law

18

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In Chapter 13, we presented the second law of thermodynamics in various different forms. In Chapter 14, we related this to the concept of entropy and showed that the entropy of an isolated system always either stays the same or increases with time. But what value does the entropy of a system take, and how can you measure it?

One way of measuring the entropy of a system is to measure its heat capacity. For example, if measurements of C_p, the heat capacity at constant pressure, are made as a function of temperature, then using

$$C_p = T\left(\frac{\partial S}{\partial T}\right)_p, \tag{18.1}$$

we can obtain entropy S by integration, so that

$$S = \int \frac{C_p}{T}\mathrm{d}T. \tag{18.2}$$

This is all very well, but when you integrate, you have to worry about constants of integration. Writing eqn 18.2 as a definite integral, we have that the entropy $S(T)$, measured at temperature T, is

$$S(T) = S(T_0) + \int_{T_0}^{T} \frac{C_p}{T}\mathrm{d}T, \tag{18.3}$$

where T_0 is some different temperature (see Fig. 18.1). Thus it seems that we are only able to learn about *changes* in entropy, for example as a system is warmed from T_0 to T, and we are not able to obtain an absolute measurement of entropy itself. The third law of thermodynamics, presented in this chapter, gives us additional information because it provides a value for the entropy at one particular temperature, namely absolute zero.

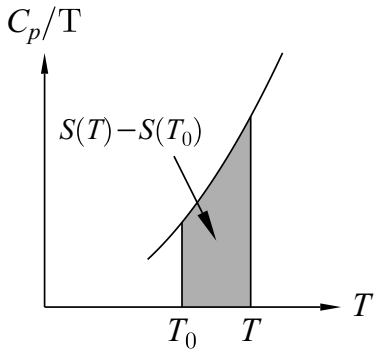


Fig. 18.1 A graphical representation of eqn 18.3.

18.1 Different statements of the third law

Walter H. Nernst (1864–1941) (Fig. 18.2) came up with the first statement of the third law of thermodynamics after examining data on chemical thermodynamics and doing experiments with electrochemical cells. The essential conclusion he came to concerned the change in enthalpy ΔH in a reaction (the heat of the reaction, positive if endothermic, negative if exothermic; see Section 16.5), and the change in Gibbs' function ΔG (which determines in which direction the reaction goes). Since

$G = H - TS$, we expect that

$$\Delta G = \Delta H - T\Delta S, \tag{18.4}$$

so that as $T \to 0$, $\Delta G \to \Delta H$. Experimental data showed that this was true, but ΔG and ΔH not only came closer together on cooling, but they approached each other asymptotically. On the basis of the data, Nernst also postulated that $\Delta S \to 0$ as $T \to 0$. His statement of the third law, dating from 1906, can be written as

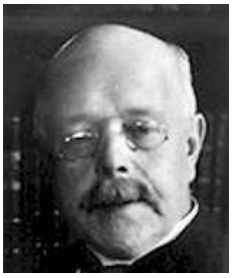


Fig. 18.2 W. Nernst

Nernst's statement of the third law
Near absolute zero, all reactions in a system in internal equilibrium take place with no change in entropy.

Max Planck (1858–1947) (Fig. 18.3) added more meat to the bones of the statement by making a further hypothesis in 1911, namely that:



Fig. 18.3 M. Planck

Planck's statement of the third law
The entropy of all systems in internal equilibrium is the same at absolute zero, and may be taken to be zero.

Planck actually made his statement only about perfect crystals. However, it is believed to be true about any system, as long as it is in internal equilibrium (i.e. that all parts of a system are in equilibrium with each other). There are a number of systems, such as ^{4}He and ^{3}He, which are liquids even at very low temperature. Electrons in a metal can be treated as a gas all the way down to $T = 0$. The third law applies to all of these systems. However, note that the systems have to be in *internal equilibrium* for the third law to apply. An example of a system not in equilibrium is a glass, which has frozen-in disorder. For a solid, the lowest–energy phase is the perfect crystal, but the glass phase is higher in energy and is unstable. The glass phase will eventually relax back to the perfect crystalline phase but it may take many years or centuries to do this.

Planck's choice of zero for the entropy was further motivated by the development of statistical mechanics, a subject we will tackle later in this book. It suffices to say here that the statistical definition of entropy, presented in eqn 14.36 ($S = k_B \ln \Omega$), implies that zero entropy is equivalent to $\Omega = 1$. Thus at absolute zero, when a system finds its ground state, the entropy being equal to zero implies that this ground state is non-degenerate.

At this point, we can raise a potential objection to the third law in Planck's form. Consider a perfect crystal composed of N spinless atoms. We are told by the third law that its entropy is zero. However, let us further suppose that each atom has at its centre a nucleus with angular momentum quantum number I. If no magnetic field is applied to this system, then we appear to have a contradiction. The degeneracy of the

nuclear spin is $2I + 1$ and if $I > 0$, this will not be equal to one. How can we reconcile this with zero entropy since the non-zero nuclear spin implies that the entropy S of this system should be $S = Nk_{\mathrm{B}} \ln(2I+1)$, to however low a temperature we cool it?

The answer to this apparent contradiction is as follows: in a real system in internal equilibrium, the individual components of the system must be able to exchange energy with each other, i.e. to *interact* with each other. Nuclear spins actually feel a tiny, but non-zero, magnetic field due to the dipolar fields produced each other, and this lifts the degeneracy. Another way of looking at this is to say that the interactions give rise to collective excitations of the nuclear spins. These collective excitations are nuclear spin waves, and the lowest–energy nuclear spin wave, corresponding to the longest–wavelength mode, will be non-degenerate. At sufficiently low temperatures (and this will be extremely low!) only that long-wavelength mode will be thermally occupied and the entropy of the nuclear spin system will be zero.

However, this example raises an important point. If we cool a crystal, we will extract energy from the lattice and its entropy will drop towards zero. However, the nuclear spins will still retain their entropy until cooled to a much lower temperature (reflecting the weaker interactions between nuclear spins compared to the bonds between atoms in the lattice). If we find a method of cooling the nuclei, there might still be some residual entropy associated with the individual nucleons. All these thermodynamic subsystems (the electrons, the nuclear spins, and the nucleons) are very weakly coupled to each other, but their entropies are additive. Francis Simon (1893–1956) (Fig. 18.4) in 1937 called these different subsystems 'aspects' and formulated the third law as follows:



Fig. 18.4 F. E. Simon

Simon's statement of the third law
The contribution to the entropy of a system by each aspect of the system which is in internal thermodynamic equilibrium tends to zero as $T \to 0$.

Simon's statement is convenient because it allows us to focus on a particular aspect of interest, knowing that its entropy will tend to zero as T approaches 0, while ignoring the aspects that we don't care about and which might not lose their entropy until much closer to $T = 0$.

18.2 Consequences of the third law

Having provided various statements of the third law, it is time to examine some of its consequences.

- *Heat capacities tend to zero as* $T \to 0$
  This consequence is easy to prove. Any heat capacity C given by

$$C = T\left(\frac{\partial S}{\partial T}\right) = \left(\frac{\partial S}{\partial \ln T}\right) \to 0, \tag{18.5}$$

because as $T \to 0$, $\ln T \to -\infty$ and $S \to 0$. Hence $C \to 0$.
Note that this result disagrees with the classical prediction of $C = R/2$ per mole per degree of freedom. (We note for future reference that this observation emphasizes the fact that the equipartition theorem, to be presented in Chapter 19, is a high temperature theory and fails at low temperature.)

- *Thermal expansion stops*
  Since $S \to 0$ as $T \to 0$, we have for example that

$$\left(\frac{\partial S}{\partial p}\right)_T \to 0 \tag{18.6}$$

  as $T \to 0$, but by a Maxwell relation, this implies that

$$\frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p \to 0 \tag{18.7}$$

  and hence the isobaric expansivity $\beta_p \to 0$.

- *No gases remain ideal as* $T \to 0$
  The ideal monatomic gas has served us well in this book as a simple model which allows us to obtain tractable results. One of these results is eqn 11.26, which states that for an ideal gas, $C_p - C_V = R$ per mole. However, as $T \to 0$, both C_p and C_V tend to zero, and this equation cannot be satisfied. Moreover, we expect that $C_V = 3R/2$ per mole, and as we have seen, this also does not work down to absolute zero. Yet another nail in the coffin of the ideal gas is the expression for its entropy given in eqn 16.82 ($S = C_V \ln T + R \ln V + \text{constant}$). As $T \to 0$, this equation yields $S \to -\infty$, which is as far from zero as you can get!
  Thus we see that the third law forces us to abandon the ideal gas model when thinking about gases at low temperature. Of course, it is at low temperature that the weak interactions between gas molecules (blissfully neglected so far since we have modelled gas molecules as independent entities) become more important. More sophisticated models of gases will be considered in Chapter 26.

- *Curie's law breaks down*
  Curie's law states that the susceptibility χ is proportional to $1/T$ and hence $\chi \to \infty$ as $T \to 0$. However, the third law implies that $(\partial S/\partial B)_T \to 0$ and hence

$$\left(\frac{\partial S}{\partial B}\right)_T = \left(\frac{\partial m}{\partial T}\right)_B = \frac{VB}{\mu_0}\left(\frac{\partial \chi}{\partial T}\right)_B \tag{18.8}$$

  must tend to zero. Thus $\left(\frac{\partial \chi}{\partial T}\right) \to 0$, in disagreement with Curie's law. Why does it break down? You may begin to see a theme developing: it is interactions again! Curie's law is derived by considering magnetic moments to be entirely independent, in which case their properties can be determined by considering only the balance between the applied field (driving the moments to align)

and temperature (driving the moments to randomize). The susceptibility measures their infinitesimal response to an infinitesimal applied field; this becomes infinite when the thermal fluctuations are removed at $T = 0$. However, if interactions between the magnetic moments are switched on, then an applied field will have much less of an effect because the magnetic moments will already be driven into some partially ordered state by each other.
There is a basic underlying message here: the microscopic parts of a system can behave independently at high temperature, where the thermal energy $k_{\mathrm{B}}T$ is much larger than any interaction energy. At low temperature, these interactions become important and all notions of independence break down. To paraphrase (badly) the poet John Donne:

No man is an island, and especially not as $T \to 0$.

- *Unattainability of absolute zero*
The final point can almost be elevated to the status of another statement of the third law:

It is impossible to cool to $T = 0$ in a finite number of steps.

This is messy to prove rigorously, but we can justify the argument by reference to Fig. 18.5, which shows plots of S against T for different values of a parameter X (which might be magnetic field, for example). Cooling is produced by isothermal increases in X and adiabatic decreases in X. If the third law did not hold, it would be possible to proceed according to Fig. 18.5(a) and cool all the way to absolute zero. However, because of the third law, the situation is as in Fig. 18.5(b) and the number of steps needed to get to absolute zero becomes infinite.

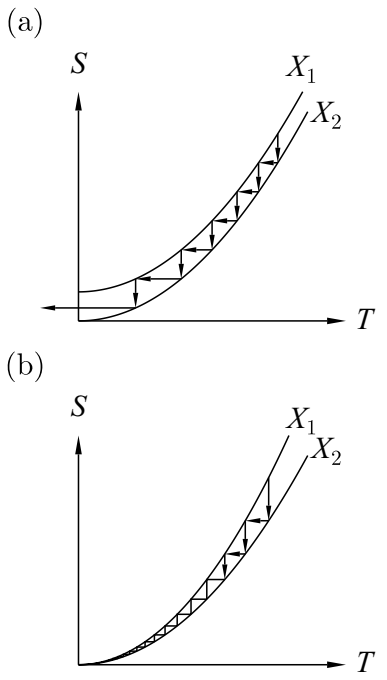


Fig. 18.5 The entropy as a function of temperature for two different values of a parameter X. Cooling is produced by isothermal increases in X (i.e. $X_1 \to X_2$) and adiabatic decreases in X (i.e. $X_2 \to X_1$). (a) If S does not go to 0 as $T \to 0$ it is possible to cool to absolute zero in a finite number of steps. (b) If the third law is obeyed, then it is impossible to cool to absolute zero in a finite number of steps.

Before concluding this chapter, we make one remark concerning Carnot engines. Consider a Carnot engine, operating between reservoirs with temperatures T_ℓ and T_{h}, having an efficiency which is equal to $\eta = 1 - (T_\ell/T_{\mathrm{h}})$ (eqn 13.10). If $T_\ell \to 0$, the efficiency η tends to 1. If you operated this Carnot engine, you would then get perfect conversion of heat into work, in violation of Kelvin's statement of the second law of thermodynamics. It seems at first sight that the unattainability of absolute zero (a version of the third law) is a simple consequence of the second law. However, there are difficulties in considering a Carnot engine operating between two reservoirs, one of which is at absolute zero. It is not clear how you can perform an isothermal process at absolute zero, because once a system is at absolute zero it is not possible to get it to change its thermodynamical state without warming it. Thus it is generally believed that the third law is indeed a separate postulate which is independent of the second law. The third law points to the fact that many of our 'simple' thermodynamic models, such as the ideal gas equation and Curie's law of paramagnets, need substantial modification

Chapter summary

- The equipartition theorem states that if the energy of a system is the sum of n quadratic modes, and that the system is in contact with a heat reservoir of temperature T, the mean energy of the system is given by $n \times \frac{1}{2}k_{\mathrm{B}}T$.
- The equipartition theorem is a high–temperature result and gives incorrect predictions at low temperature, where the discrete nature of the energy spectrum cannot be ignored.

Exercises

(19.1) What is the mean kinetic energy in eV at room temperature of a gaseous (a) He atom, (b) Xe atom, (c) Ar atom and (d) Kr atom. [Hint: do you have to do four separate calculations?]

(19.2) Comment on the following values of molar heat capacity in $\mathrm{J\,K^{-1}\,mol^{-1}}$, all measured at constant pressure at 298 K.

Al	24.35	Pb	26.44
Ar	20.79	Ne	20.79
Au	25.42	N_2	29.13
Cu	24.44	O_2	29.36
He	20.79	Ag	25.53
H_2	28.82	Xe	20.79
Fe	25.10	Zn	25.40

[Hint: express them in terms of R; which of the substances is a solid and which is gaseous?]

(19.3) A particle at position r is in a potential well $V(r)$ given by

$$V(r) = \frac{A}{r^n} - \frac{B}{r}, \tag{19.28}$$

where A and B are positive constants and $n > 2$. Show that the bottom of the well is approximately quadratic in r. Hence find the particle's mean thermal energy at temperature T above the bottom of the well assuming the validity of the equipartition theorem in this situation.

(19.4) In example 19.1, show that

$$\langle x_i^2 \rangle = \frac{k_{\mathrm{B}}T}{2\alpha_i}. \tag{19.29}$$

(19.5) If the energy E of a system is not quadratic, but behaves like $E = \alpha|x|$ where $\alpha > 0$, show that $\langle E \rangle = k_{\mathrm{B}}T$.

(19.6) If the energy E of a system behaves like $E = \alpha|x|^n$, where $n = 1, 2, 3\ldots$ and $\alpha > 0$, show that $\langle E \rangle = \xi k_{\mathrm{B}}T$, where ξ is a numerical constant.

(19.7) A simple pendulum with length ℓ makes an angle θ with the vertical, where $\theta \ll 1$. Show that it oscillates with a period given by $2\pi\sqrt{\ell/g}$. The pendulum is now placed at rest and allowed to come into equilibrium with its surroundings at temperature T. Derive an expression for $\langle \theta^2 \rangle$.

The partition function

20

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The probability that a system is in some particular state α is given by the Boltzmann factor $\mathrm{e}^{-\beta E_\alpha}$. We define the **partition function**[1] Z by a sum over all the states of the Boltzmann factors, so that

$$\boxed{Z = \sum_\alpha \mathrm{e}^{-\beta E_\alpha}} \tag{20.1}$$

where the sum is over all states of the system (each one labelled by α). The partition function Z contains all the information about the energies of the states of the system, and the fantastic thing about the partition function is that all thermodynamical quantities can be obtained from it. It behaves like a zipped-up and compressed version of all the properties of the system; once you have Z, you only have to know how to uncompress and unzip it to get functions of state like energy, entropy, Helmholtz function, or heat capacity to simply drop out. We can therefore reduce problem-solving in statistical mechanics to two steps:

Steps to solving statistical mechanics problems:

(1) Write down the partition function Z.
(see Section 20.1)

(2) Go through some standard procedures to obtain the functions of state you want from Z.
(see Section 20.2)

We will outline these two steps in the sections that follow. Before we do that, let us pause to notice an important feature about the partition function.

- The zero of energy is always somewhat arbitrary: one can always choose to measure energy with respect to a different zero, since it is only energy differences which are important. Hence the partition function is defined up to an arbitary multiplicative constant. This seems somewhat strange, but it turns out that many physical quantities are related to the logarithm of the partition function and therefore these quantities are defined up to an additive constant (which might reflect, for example, the rest mass of particles). Other physical quantities, however, are determined by a differential of the logarithm of the partition function and therefore these quantities can be determined precisely.

[1] The partition function is given the symbol Z because the concept was first coined in German. *Zustandssumme* means 'sum over states', which is exactly what Z is. The English name 'partition function' reflects the way in which Z measures how energy is 'partitioned' between states of the system.

Cooling real gases

27

In Chapter 26, we considered how to model the properties of real gases using various corrections to the ideal gas model. In this chapter, we will use these results to explore some of the deviations from ideal gas behaviour which can be observed in practice, in particular with changes in the behaviour of a Joule expansion. Then we will introduce the Joule–Kelvin throttling process (which has no effect on an ideal gas, but which can lead to cooling of a real gas) and discuss how real gases can be liquefied.

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27.1 The Joule expansion

We have discussed the properties of non-ideal gases in some detail. In this section, we will see how the intermolecular interactions in such gases lead to departures from ideal-gas behaviour for the Joule expansion. Recall from Section 14.4 that a Joule expansion is an irreversible expansion of a gas into a vacuum which can be accomplished by opening a tap connecting the vessel containing gas and an evacuated vessel (see Fig. 27.1). The entire system is isolated from its surroundings and so no heat enters or leaves. No work is done, so the internal energy U is unchanged. We are interested in finding out whether the gas warms, cools or remains at constant temperature in this expansion.

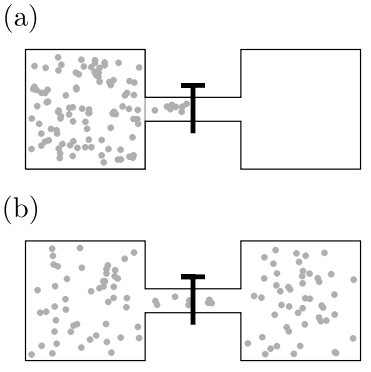


Fig. 27.1 The Joule expansion: (a) before opening the tap and (b) after opening the tap.

To answer this, we define the **Joule coefficient** μ_{J} using

$$\mu_{\mathrm{J}} = \left(\frac{\partial T}{\partial V}\right)_U, \tag{27.1}$$

where the constraint of constant U is relevant for the Joule expansion. This partial differential can be transformed using eqn 16.67 and the definition of C_V to give

$$\mu_{\mathrm{J}} = -\left(\frac{\partial T}{\partial U}\right)_V \left(\frac{\partial U}{\partial V}\right)_T = -\frac{1}{C_V}\left(\frac{\partial U}{\partial V}\right)_T. \tag{27.2}$$

Now the first law, $\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$, implies that

$$\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial S}{\partial V}\right)_T - p, \tag{27.3}$$

and using a Maxwell relation (eqn 16.53) this becomes

$$\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial p}{\partial T}\right)_V - p, \tag{27.4}$$

and hence

$$\mu_{\mathrm{J}} = -\frac{1}{C_V}\left[T\left(\frac{\partial p}{\partial T}\right)_V - p\right]. \tag{27.5}$$

For an ideal gas, $p = RT/V$, $(\partial p/\partial T)_V = R/V$, and hence $\mu_{\mathrm{J}} = 0$. Hence, as we found in Section 14.4, the temperature of an ideal gas is unchanged in a Joule expansion. For real gases, you always get cooling because of the attractive effect of interactions. This is because $C_V > 0$ and $(\partial U/\partial V)_T > 0$ and so $\mu_{\mathrm{J}} = -\frac{1}{C_V}(\partial U/\partial V)_T < 0$. This can be understood physically as follows: when a gas freely expands into a vacuum, the time-averaged distance between neighbouring molecules increases and the magnitude of the potential energy resulting from the attractive intermolecular interactions is reduced. However, this potential energy is a negative quantity (because the interactions are attractive) and so the potential energy is actually *increased* (because it is made less negative).[1] Since U must be unchanged in a Joule expansion (no heat enters or leaves and no work is done), the kinetic energy must be reduced (by the same amount by which the potential energy rises) and hence the temperature falls.

[1] Of course, at very high densities, the intermolecular interactions become repulsive rather than attractive, but at such a density one is probably dealing with a solid rather than a gas.

Example 27.1

Evaluate the Joule coefficient for a van der Waals gas.
Solution:
The equation of state is $p = RT/(V-b) - a/V^2$ and so

$$\left(\frac{\partial p}{\partial T}\right)_V = \frac{R}{V-b}, \tag{27.6}$$

and hence

$$\mu_{\mathrm{J}} = -\frac{1}{C_V}\left[\frac{RT}{V-b} - \frac{RT}{V-b} + \frac{a}{V^2}\right] = -\frac{a}{C_V V^2}. \tag{27.7}$$

The temperature change in a Joule expansion from V_1 to V_2 can be evaluated simply by integrating the Joule coefficient as follows:

$$\Delta T = \int_{V_1}^{V_2} \mu_{\mathrm{J}}\,\mathrm{d}V = -\int_{V_1}^{V_2} \frac{1}{C_V}\left[T\left(\frac{\partial p}{\partial T}\right)_V - p\right]\mathrm{d}V. \tag{27.8}$$

Example 27.2

Evaluate the change in temperature for a van der Waals gas which undergoes a Joule expansion from volume V_1 to volume V_2.
Solution:
Using eqn 27.8, we have that

$$\Delta T = -\frac{a}{C_V}\int_{V_1}^{V_2}\frac{\mathrm{d}V}{V^2} = -\frac{a}{C_V}\left(\frac{1}{V_1} - \frac{1}{V_2}\right) < 0 \tag{27.9}$$

since $V_2 > V_1$ in an expansion.

27.2 Isothermal expansion

Consider the isothermal expansion of a non-ideal gas. Equation 27.4 states that

$$\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial p}{\partial T}\right)_V - p, \tag{27.10}$$

so that the change of U in an isothermal expansion is

$$\Delta U = \int_{V_1}^{V_2} \left[T\left(\frac{\partial p}{\partial T}\right)_V - p\right] \mathrm{d}V. \tag{27.11}$$

- For an ideal gas[2], $\Delta U = 0$.
- For a van der Waals gas, $\Delta U = \int_{V_1}^{V_2} \frac{a}{V^2}\,\mathrm{d}V = a(1/V_1 - 1/V_2)$.

[2] For an ideal gas, $p = RT/V$, $(\partial p/\partial T)_V = R/V$, and hence $T\left(\frac{\partial p}{\partial T}\right)_V - p = 0$.

Note that U depends on a, not b (it is influenced by the intermolecular interactions but does not 'care' that they have non-zero size). Note also that for large volumes, U becomes independent of V and one recovers the ideal gas limit.

Example 27.3

Calculate the entropy of a van der Waals gas.
Solution:
The entropy S can be written as a function of T and V so that $S = S(T, V)$. Hence

$$\begin{aligned} \mathrm{d}S &= \left(\frac{\partial S}{\partial T}\right)_V \mathrm{d}T + \left(\frac{\partial S}{\partial V}\right)_T \mathrm{d}V \\ &= \frac{C_V}{T}\mathrm{d}T + \left(\frac{\partial p}{\partial T}\right)_V \mathrm{d}V, \end{aligned} \tag{27.12}$$

where eqns 16.68 and 16.53 have been used to obtain the second line. For the van der Waals gas, we can write $(\partial p/\partial T)_V = R/(V - b)$, and hence we have that

$$S = C_V \ln T + R\ln(V - b) + \text{constant}. \tag{27.13}$$

Note that the entropy depends on the constant b, but not a. Entropy 'cares' about the volume occupied by the molecules in the gas (because this determines how much available space there is for the molecules to move around in, and this in turn determines the number of possible microstates of the system) but not about the intermolecular interactions.

27.3 Joule–Kelvin expansion

The Joule expansion is a useful conceptual process, but it is not much practical use for cooling gases. Gas slightly cools when it is expanded into a second evacuated vessel, but what do you do with it then? What is wanted is some kind of flow process where warm gas can be fed into some kind of a 'cooling machine' and cold gas (or better still, cold liquid) emerges from the other end. Such a process was discovered by James Joule and William Thomson (later Lord Kelvin) and is known as a **Joule–Thomson expansion** or a **Joule–Kelvin expansion**.

Consider a steady flow process in which gas at high pressure p_1 is forced through a throttle valve or a porous plug to a lower pressure p_2. This is illustrated in Fig. 27.2. Consider a volume V_1 of gas on the high–pressure side. Its internal energy is U_1. To push the gas through the constriction, the high pressure gas behind it has to do work on it equal to p_1V_1 (since the pressure p_1 is maintained on the high–pressure side of the constriction). The gas expands as it passes through to the low–pressure region and now occupies volume V_2 which is larger than V_1. It has to do work on the low–pressure gas in front of it which is at pressure p_2 and hence this work is p_2V_2. The gas may change its temperature in the process and hence its new internal energy is U_2. The change in internal energy $(U_2 - U_1)$ must be equal to the work done on the gas (p_1V_1) minus the work done by the gas (p_2V_2). Thus

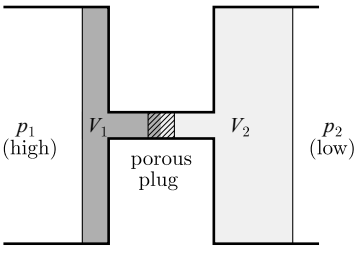


Fig. 27.2 A throttling process.

$$U_1 + p_1V_1 = U_2 + p_2V_2 \tag{27.14}$$

or equivalently

$$H_1 = H_2, \tag{27.15}$$

so that it is *enthalpy* that is conserved in this flow process.

Since we are now interested in how much the gas changes temperature when we reduce its pressure at constant enthalpy, we define the Joule–Kelvin coefficient by

$$\mu_{\mathrm{JK}} = \left(\frac{\partial T}{\partial p}\right)_H. \tag{27.16}$$

This can be transformed using the reciprocity theorem (eqn C.42) and the definition of C_p to give

$$\mu_{\mathrm{JK}} = -\left(\frac{\partial T}{\partial H}\right)_p \left(\frac{\partial H}{\partial p}\right)_T = -\frac{1}{C_p}\left(\frac{\partial H}{\partial p}\right)_T. \tag{27.17}$$

Now the relation $\mathrm{d}H = T\mathrm{d}S + V\mathrm{d}p$ implies that

$$\left(\frac{\partial H}{\partial p}\right)_T = T\left(\frac{\partial S}{\partial p}\right)_T + V, \tag{27.18}$$

and using a Maxwell relation (eqn 16.54) this becomes

$$\left(\frac{\partial H}{\partial p}\right)_T = -T\left(\frac{\partial V}{\partial T}\right)_p + V, \tag{27.19}$$

and hence

$$\mu_{\rm JK} = \frac{1}{C_p}\left[T\left(\frac{\partial V}{\partial T}\right)_p - V\right] \tag{27.20}$$

The change in temperature for a gas following a Joule–Kelvin expansion from pressure p_1 to pressure p_2 is given by

$$\Delta T = \int_{p_1}^{p_2} \frac{1}{C_p}\left[T\left(\frac{\partial V}{\partial T}\right)_p - V\right] {\rm d}p. \tag{27.21}$$

Since ${\rm d}H = T{\rm d}S + V{\rm d}p = 0$, the entropy change is

$$\Delta S = -\int_{p_1}^{p_2} \frac{V}{T}{\rm d}p, \tag{27.22}$$

and for an ideal gas this is $R\ln(p_1/p_2) > 0$. Thus this is an irreversible process.

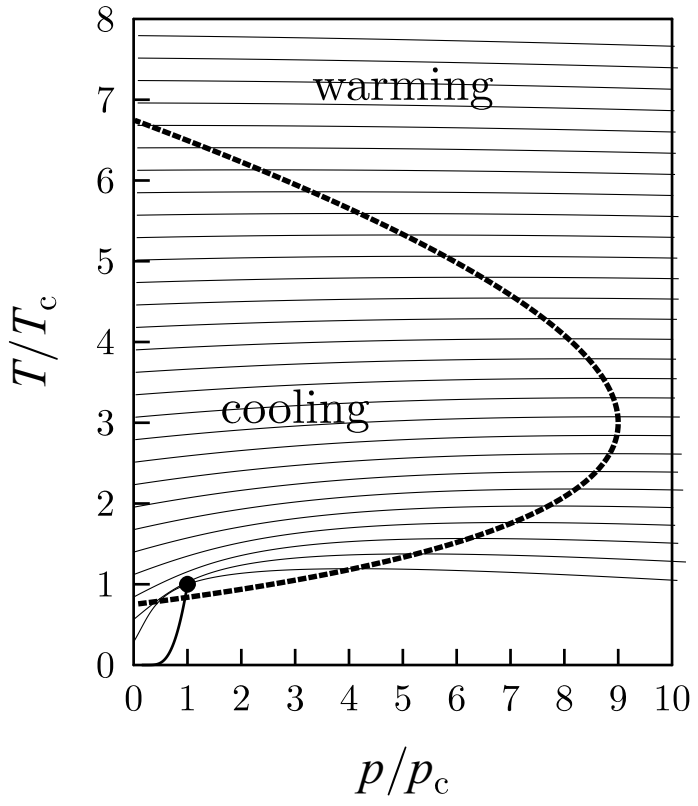


Fig. 27.3 The inversion curve of the van der Waals gas is shown as the heavy dashed line. The isenthalps (lines of constant enthalpy) are shown as thin solid lines. When the gradients of the isenthalps on this diagram are positive, then cooling can be obtained when pressure is reduced at constant enthalpy (i.e. in a Joule–Kelvin expansion). Also shown (as a solid line near the bottom left-hand corner of the graph which terminates at the dot) is the line of coexistence of liquid and gas (from Fig. 26.6) ending in the critical point ($p = p_{\rm c}$, $T = T_{\rm c}$, shown by the dot).

Whether the Joule–Kelvin expansion results in heating or cooling is more subtle and in fact $\mu_{\rm JK}$ can take either sign. It is convenient to consider when $\mu_{\rm JK}$ changes sign, and this will occur when $\mu_{\rm JK} = 0$, i.e. when $T(\partial V/\partial T)_p - V = 0$, or equivalently

$$\left(\frac{\partial V}{\partial T}\right)_p = \frac{V}{T}. \tag{27.23}$$

^{4}He	H_2	N_2	Ar	CO_2
43	204	607	794	1275

Table 27.1 The maximum inversion temperature in Kelvin for several gases.

This equation defines the so-called **inversion curve** in the T–p plane. This is plotted for the van der Waals gas in Fig. 27.3 as a heavy solid line. The lines of constant enthalpy are also shown and their gradients change sign when they cross the inversion curve. When the gradient of the isenthalps on this diagram are positive, then cooling can be obtained when pressure is reduced at constant enthalpy (i.e. in a Joule–Kelvin expansion).

A crucial parameter is the maximum inversion temperature, below which the Joule–Kelvin expansion can result in cooling. These are listed for several real gases in Table 27.1. In the case of helium, this temperature is 43 K, so helium gas must be cooled to below this temperature by some other means before it can be liquefied using the Joule–Kelvin process.

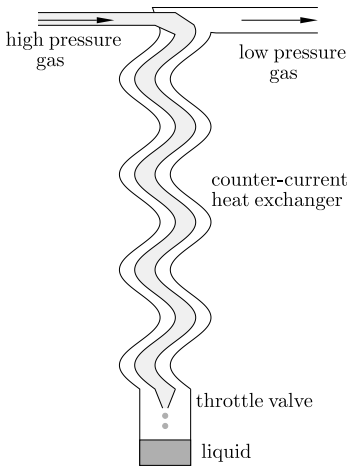


Fig. 27.4 A schematic diagram of a liquefier.

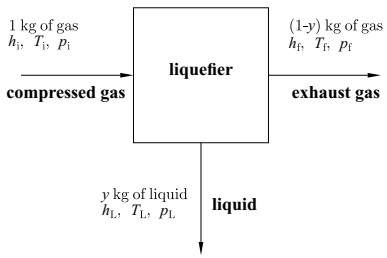


Fig. 27.5 A block diagram of the liquefaction process.

27.4 Liquefaction of gases

For achieving the liquefaction of gases, the Joule–Kelvin process is extremely useful, though it must be carried out below the maximum inversion temperature of the particular gas in question. A schematic diagram of a liquefier is shown in Fig. 27.4. High–pressure gas is forced through a throttle valve, resulting in cooling by the Joule–Kelvin process. Low–pressure gas plus liquid results, and the process is made more efficient by use of a counter-current heat exchanger by which the outgoing cold low–pressure gas is used to precool the incoming warm high pressure gas, helping to ensure that by the time it reaches the throttle valve the incoming high–pressure gas is already as cool as possible and at least at a temperature such that the Joule–Kelvin effect will result in cooling.

We can consider the liquefier as a 'black box' into which you put 1 kg of warm gas and get out y kg of liquid, as well as $(1-y)$ kg of exhaust gas (see Fig. 27.5). The variable y is the efficiency y of a liquefier, i.e. the mass fraction of incoming gas which is liquefied. Since enthalpy is conserved in a Joule–Kelvin process, we have that

$$h_\mathrm{i} = yh_\mathrm{L} + (1-y)h_\mathrm{f}, \tag{27.24}$$

where h_i is the specific enthalpy of the incoming gas, h_L is the specific enthalpy of the liquid, and h_f is the specific enthalpy of the outgoing gas. Hence the efficiency y is given by

$$y = \frac{h_\mathrm{f} - h_\mathrm{i}}{h_\mathrm{f} - h_\mathrm{L}} \tag{27.25}$$

For an efficient heat exchanger, the temperature of the compressed gas T_i and the exhaust gas T_f will be the same. We also have that $p_\mathrm{f} = 1\,\mathrm{atm}$, and T_L is fixed (because the liquid will be in equilibrium with its vapour). Therefore h_f and h_L are fixed. The only parameter to vary is then h_i and to maximize y we must minimize h_i, i.e.

$$\left(\frac{\partial h_\mathrm{i}}{\partial p_\mathrm{i}}\right)_{T_\mathrm{i}} = 0 \tag{27.26}$$

and since $(\partial h/\partial p)_T = -(1/C_p)\mu_\mathrm{JK}$, we therefore require that

$$\mu_\mathrm{JK} = 0. \tag{27.27}$$

This means that it is best to work the liquefier right on the inversion curve ($\mu_\mathrm{JK} = 0$) for maximum efficiency.

Most gases could be liquefied by the end of the nineteenth century, but the modern form of gas liquefier dates back to the work of the German chemist Karl von Linde (1842–1934) who commercialized liquid-air production in 1895 using the Joule–Kelvin effect with a counter-current heat exchanger (as shown in Fig. 27.4; this is known as the **Linde process**) and discovered various uses of liquid nitrogen. James Dewar (1842–1923) was the first to liquefy hydrogen using the Linde process in 1898, and in 1899 got it to go solid. Dewar was also the first, in 1891, to study the magnetic properties of liquid oxygen. The Dutch physicist Heike Kamerlingh Onnes (1853–1926) was the first to produce liquid helium in 1908 by a similar process, precooling the helium gas using liquid hydrogen. Using liquid helium, he then discovered superconductivity in 1911, and was awarded the Nobel Prize in 1913 for 'his investigations on the properties of matter at low temperatures which led, inter alia, to the production of liquid helium'.

Chapter summary

- The Joule expansion results in cooling for non-ideal gases because of the attractive interactions between molecules.
- The entropy of a gas depends on the non-zero size of molecules.
- The Joule–Kelvin expansion is a steady flow process in which enthalpy is conserved. It can result in either warming or cooling of a gas. It forms the basis of many gas liquefaction techniques.

Exercises

(27.1) (a) Derive the following general relations

$$\text{(a)} \quad \left(\frac{\partial T}{\partial V}\right)_U = -\frac{1}{C_V}\left[T\left(\frac{\partial p}{\partial T}\right)_V - p\right],$$

$$\text{(b)} \quad \left(\frac{\partial T}{\partial V}\right)_S = -\frac{1}{C_V}T\left(\frac{\partial p}{\partial T}\right)_V,$$

$$\text{(c)} \quad \left(\frac{\partial T}{\partial p}\right)_H = \frac{1}{C_p}\left[T\left(\frac{\partial V}{\partial T}\right)_p - V\right].$$

In each case the quantity on the left hand side is the appropriate thing to consider for a particular type of expansion. State what type of expansion each refers to.
(b) Using these relations, verify that for an ideal gas $(\partial T/\partial V)_U = 0$ and $(\partial T/\partial p)_H = 0$, and that $(\partial T/\partial V)_S$ leads to the familiar relation $pV^\gamma =$ constant along an isentrope.

(27.2) In a Joule–Kelvin liquefier, gas is cooled by expansion through an insulated throttle – a simple but inefficient process with no moving parts at low temperature. Explain why enthalpy is conserved in this process. Deduce that

$$\left(\frac{\partial T}{\partial P}\right)_H = \frac{1}{C_P}\left[T\left(\frac{\partial V}{\partial T}\right)_P - V\right].$$

Estimate the highest starting temperature at which the process will work for helium at low densities, on the following assumptions:
(i) the pressure is given at low densities by a virial expansion of the form

$$\frac{PV}{RT} = 1 + \left(b - \frac{a}{RT}\right)\left(\frac{1}{V}\right) + \cdots,$$

and
(ii) the Boyle temperature a/bR (the temperature at which the second virial coefficient vanishes) is known from experiment to be 19 K for helium.

[Hint: One method of solving this problem is to remember that p is easily made the subject of the equation of state and one can then use $(\partial V/\partial T)_p = -(\partial p/\partial T)_V/(\partial p/\partial V)_T$.]

(27.3) For a gas obeying Dieterici's equation of state

$$p(V-b) = RT\mathrm{e}^{-a/RTV},$$

for 1 mole, prove that the equation of the inversion curve is

$$p = \left(\frac{2a}{b^2} - \frac{RT}{b}\right)\exp\left(\frac{1}{2} - \frac{a}{RTb}\right),$$

and hence find the maximum inversion temperature $T_{\max}$.

(27.4) Show that the equation for the inversion curve of the Dieterici gas in reduced units is

$$\tilde{P} = (8 - \tilde{T})\exp\left[\frac{5}{2} - \frac{4}{\tilde{T}}\right],$$

and sketch it in the $\tilde{T}$–$\tilde{P}$ plane.

(27.5) Why is enthalpy conserved in steady flow processes? A helium liquefier in its final stage of liquefaction takes in compressed helium gas at 14 K, liquefies a fraction α, and rejects the rest at 14 K and atmospheric pressure. Use the values of enthalpy H of helium gas at 14 K as a function of pressure p in the table below to determine the input pressure which allows α to take its maximum value, and determine what this value is.

p (atm)	0	10	20	30	40
H (kJ kg^{-1})	87.4	78.5	73.1	71.8	72.6

[Enthalpy of liquid helium at atmospheric pressure = 10.1 kJ kg^{-1}].

Phase transitions

28

In this chapter we will consider **phase transitions**, in which one thermodynamic phase changes into another. An example would be the transition from liquid water to gaseous steam which occurs when you boil a kettle of water. If you start with cold water, and warm it in the kettle, all that happens initially is that the water gets progressively hotter. However, when the temperature of the water reaches 100°C interesting things begin to happen. Bubbles of gas form with different sizes, making the kettle considerably noisier, and water molecules begin to leave the liquid surface in large quantities and steam is emitted. The transition between different phases is very sudden. It is only when the boiling point is reached that liquid water becomes thermodynamically unstable and gaseous water, steam, becomes thermodynamically stable. In this chapter, we will look in detail at the thermodynamics of this and other phase transitions.

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28.1 Latent heat

To increase the temperature of a substance, one needs to apply heat, and how much heat is needed can be calculated from the heat capacity because adding heat to the substance increases its entropy. The gradient of entropy with temperature is related to the heat capacity via

$$C_x = T\left(\frac{\partial S}{\partial T}\right)_x, \tag{28.1}$$

where x is the appropriate constraint (e.g. p, V, B etc). Now consider two phases which are in thermodynamic equilibrium at a critical temperature T_c. Very often, it is found that to change from phase 1 to phase 2 at a constant temperature T_c, you need to supply some extra heat, known as the **latent heat** L, which is given by

$$L = \Delta Q_{\text{rev}} = T_c(S_2 - S_1), \tag{28.2}$$

where S_1 is the entropy of phase 1 and S_2 is the entropy of phase 2. This, together with eqn 28.1, implies that there will be a spike in the heat capacity C_x as a function of temperature.

An example of a phase transition which involves a latent heat is the liquid–gas transition. The entropy as a function of temperature for H_2O is shown in Fig. 28.1. The entropy is shown to change discontinuously at the phase transition. The heat capacity[1] C_p of the liquid phase,

[1] We use C_p because the constraint usually applied in the laboratory is that of constant pressure.

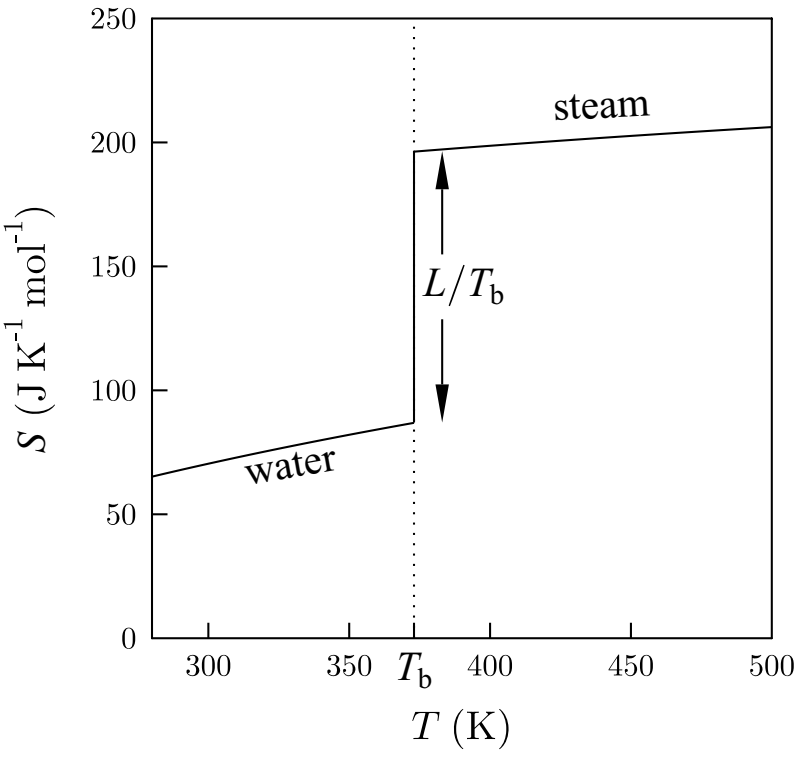


Fig. 28.1 The entropy of H_2O as a function of temperature. The boiling point is $T_b = 373\,K$.

water, is about $75\,J\,K^{-1}\,mol^{-1}$ (equivalent to about $4.2\,kJ\,kg^{-1}\,K^{-1}$) at temperatures below the boiling point T_b, and this is responsible for the gradient of $S(T)$ below the transition (because $\Delta S = \int C_p\,dT/T$), while the heat capacity of the gaseous phase, steam, is about $34\,J\,K^{-1}\,mol^{-1}$, and this is responsible for the gradient in $S(T)$ above the transition. The sudden, discontinuous change in S which occurs at T_b is a jump of magnitude L/T_b, where L is the latent heat, equal to $40.7\,kJ\,mol^{-1}$ (or equivalently $2.26\ MJ\,kg^{-1}$).

Example 28.1

If it takes 3 minutes to boil a kettle of water which was initially at 20°C, how much longer will it take to boil the kettle dry?

Solution:

Using the data above, the energy required to raise water from 20°C to 100°C is $80 \times 4.2 = 336\,kJ\,kg^{-1}$. The energy required to turn it into steam at 100°C is $2.26\ MJ\,kg^{-1}$, which is 6.7 times as big. Therefore it would take $6.7 \times 3 \approx 20$ minutes to boil the kettle dry (though, of course, having an automatic switch-off mechanism saves this from happening!).

Let us now perform a rough estimate for the entropy discontinuity at a vapour[2]–liquid transition. The number of microstates Ω available to a single gas molecule is proportional to its volume,[3] and hence the ratio of Ω for one mole of vapour and one mole of liquid is

$$\frac{\Omega_{\text{vapour}}}{\Omega_{\text{liquid}}} = \left(\frac{V_{\text{vapour}}}{V_{\text{liquid}}}\right)^{N_{\text{A}}}. \tag{28.3}$$

Hence

$$\frac{\Omega_{\text{vapour}}}{\Omega_{\text{liquid}}} = \left(\frac{\rho_{\text{liquid}}}{\rho_{\text{vapour}}}\right)^{N_{\text{A}}} \sim (10^3)^{N_{\text{A}}}, \tag{28.4}$$

since the density of the vapour is roughly 10^3 times smaller than the density of the liquid. Hence, using $S = k_{\text{B}} \ln \Omega$, we have that the entropy discontinuity is approximately

$$\Delta S = \Delta(k_{\text{B}} \ln \Omega) = k_{\text{B}} \ln(10^3)^{N_{\text{A}}} = R \ln 10^3 \approx 7R, \tag{28.5}$$

Remember that $R = N_{\text{A}} k_{\text{B}}$.

so that

$$L \approx 7RT_{\text{b}}. \tag{28.6}$$

This relationship is known as **Trouton's rule**, and is an empirical relationship which has been noticed for many systems, although it is usually stated with a slightly different prefactor:

$$L \approx 10RT_{\text{b}}. \tag{28.7}$$

The fact that the latent heat is slightly larger than expected from our simple argument stems from the fact that the latent heat also involves a contribution from the attractive intermolecular potential. However, the law of corresponding states[4] implies that if substances have similar shaped intermolecular potentials then certain properties should scale in the same way, so we do expect L/RT_{b} to be a constant.

[4] See Section 26.4.

	Ne	Ar	Kr	Xe	He	H_2O	CH_4	C_6H_6
T_{b} (K)	27.1	87.3	119.8	165.0	4.22	373.15	111.7	353.9
L (kJ mol^{-1})	1.77	6.52	9.03	12.64	0.084	40.7	8.18	30.7
L/RT_{b}	7.85	8.98	9.06	9.21	2.39	13.1	8.80	10.5

Table 28.1 The values of T_{b}, L and L/RT_{b} for several common substances.

This can be tested for various real substances (see Table 28.1) and indeed it is found that for many substances the ratio L/RT_{b} is around 8–10, confirming Trouton's empirical rule. Notable outliers include helium (He) for which quantum effects are very important (see Chapter 30) and water[5] (H_2O), which is a polar liquid (because the water molecule has a dipole moment) and which therefore possesses a rather different intermolecular potential.

[2] The word *vapour* is a synonym for gas, but is often used when conditions are such that the substance in the gas can also exist as a liquid or solid; if $T < T_{\text{c}}$, the vapour can be condensed into a liquid or solid with the application of pressure.

[3] Recall from Section 21.1 that one microstate occupies a volume in k-space equal to $(2\pi/L)^3 \propto V$, and hence the density of states is proportional to the system volume V.

[5] Water being a special case has a lot of consequences; see, for example, Section 37.3.

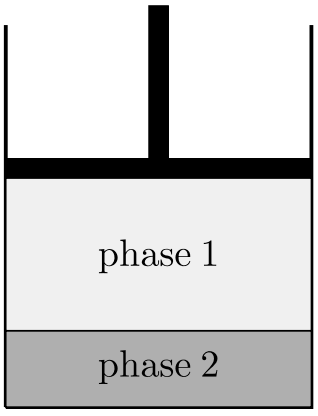


Fig. 28.2 Two phases in equilibrium at constant pressure (the constraint of constant pressure is maintained by the piston).

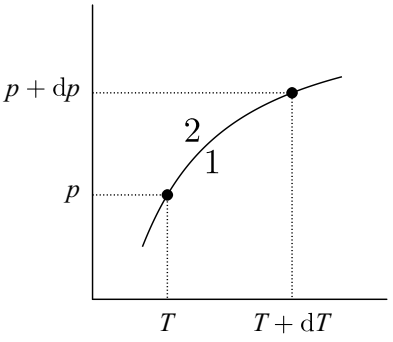


Fig. 28.3 Two phases in the p–T plane coexist at the phase boundary, shown by the solid line.

28.2 Chemical potential and phase changes

We have seen in Section 16.5 that the Gibbs function is the quantity that must be minimized when systems are held at constant pressure and temperature. In Section 22.5 we found that the chemical potential is the Gibbs function per particle. We were also able to write in eqn 22.52 that

$$\mathrm{d}G = V\mathrm{d}p - S\mathrm{d}T + \sum_i \mu_i \mathrm{d}N_i. \tag{28.8}$$

Now consider the situation in Figure 28.2 in which N_1 particles of phase 1 are in equilibrium with N_2 particles of phase 2. Then the total Gibbs free energy is

$$G_{\mathrm{tot}} = N_1\mu_1 + N_2\mu_2, \tag{28.9}$$

and since we are in equilibrium we must have

$$\mathrm{d}G_{\mathrm{tot}} = 0, \tag{28.10}$$

and hence

$$\mathrm{d}G_{\mathrm{tot}} = \mathrm{d}N_1\mu_1 + \mathrm{d}N_2\mu_2 = 0. \tag{28.11}$$

But if we increase the number of particles in phase 1, the number of particles in phase 2 must decrease by the same amount, so that $\mathrm{d}N_1 = -\mathrm{d}N_2$. Hence we have that

$$\mu_1 = \mu_2. \tag{28.12}$$

Thus in phase equilibrium, each coexisting phase has the same chemical potential. The lowest μ phase is stable. Along a line of coexistence $\mu_1 = \mu_2$.

28.3 The Clausius–Clapeyron equation

We now want to find the equation which describes the phase boundary in the p–T plane (see Fig. 28.3). This line of coexistence of the two phases is determined by the equation

$$\mu_1(p, T) = \mu_2(p, T). \tag{28.13}$$

If we move along this phase boundary, we must also have

$$\mu_1(p + \mathrm{d}p, T + \mathrm{d}T) = \mu_2(p + \mathrm{d}p, T + \mathrm{d}T), \tag{28.14}$$

so that when we change p to $p + \mathrm{d}p$ and T to $T + \mathrm{d}T$ we must have

$$\mathrm{d}\mu_1 = \mathrm{d}\mu_2. \tag{28.15}$$

This implies that (using eqns 16.22 and 22.48)

$$-s_1\mathrm{d}T + v_1\mathrm{d}p = -s_2\mathrm{d}T + v_2\mathrm{d}p, \tag{28.16}$$

where s_1 and s_2 are the entropy per particle in phases 1 and 2, and v_1 and v_2 are the volume per particle in phases 1 and 2. Rearranging this equation therefore gives that

$$\frac{\mathrm{d}p}{\mathrm{d}T} = \frac{s_2 - s_1}{v_2 - v_1}. \tag{28.17}$$

If we define the latent heat per particle as $l = T\Delta s$, we then have that

$$\frac{\mathrm{d}p}{\mathrm{d}T} = \frac{l}{T(v_2 - v_1)}, \tag{28.18}$$

or equivalently

$$\boxed{\frac{\mathrm{d}p}{\mathrm{d}T} = \frac{L}{T(V_2 - V_1).}} \tag{28.19}$$

which is known as the **Clausius–Clapeyron equation**. This shows that the gradient of the phase boundary of the p–T plane is purely determined by the latent heat, the temperature at the phase boundary and the difference in volume between the two phases.[6]

[6] This can be obtained from the difference in *densities*.

Example 28.2

Derive an equation for the phase boundary of the liquid and gas phases under the assumptions that the latent heat L is temperature independent, that the vapour can be treated as an ideal gas, and that $V_{\mathrm{vapour}} = V \gg V_{\mathrm{liquid}}$.

Solution:

Assuming that $V_{\mathrm{vapour}} = V \gg V_{\mathrm{liquid}}$ and that $pV = RT$ for one mole, the Clausius–Clapeyron equation becomes

$$\frac{\mathrm{d}p}{\mathrm{d}T} = \frac{Lp}{RT^2}. \tag{28.20}$$

This can be rearranged to give

$$\frac{\mathrm{d}p}{p} = \frac{L\mathrm{d}T}{RT^2}, \tag{28.21}$$

and hence integrating we obtain

$$\ln p = -\frac{L}{RT} + \text{constant}. \tag{28.22}$$

Hence the equation of the phase boundary is

$$p = p_0 \exp\left(-\frac{L}{RT}\right), \tag{28.23}$$

where the exponential looks like a Boltzmann factor $\mathrm{e}^{-\beta l}$ with $l = L/N_{\mathrm{A}}$, the latent heat per particle.

Remember again that $R = N_{\mathrm{A}}k_{\mathrm{B}}$.

The temperature dependence of the latent heat of many substances cannot be neglected. As an example, the temperature dependence of the latent heat of water is shown in Fig. 28.4 and this shows a weak temperature dependence. A method of treating this is outlined in the following example.

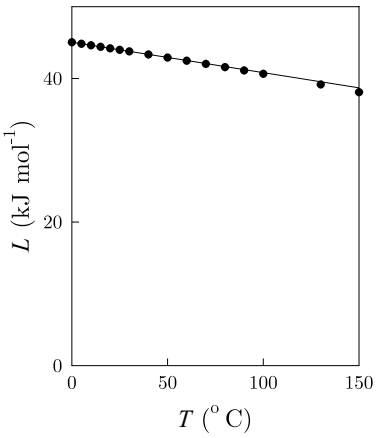


Fig. 28.4 The temperature dependence of the latent heat of water. The solid line is according to eqn 28.30.

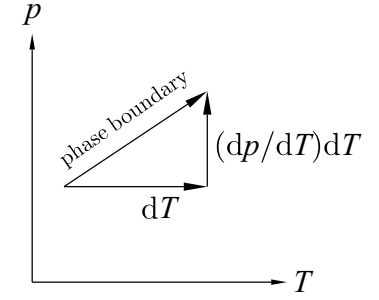


Fig. 28.5 The phase boundary.

Example 28.3

Evaluate the temperature dependence of the latent heat along the phase boundary in a liquid–gas transition and hence deduce the equation of the phase boundary including this temperature dependence.

Solution:

Along the phase boundary, we can write that the gradient in the temperature is (see Fig. 28.5)

$$\frac{\mathrm{d}}{\mathrm{d}T} = \left(\frac{\partial}{\partial T}\right)_p + \frac{\mathrm{d}p}{\mathrm{d}T}\left(\frac{\partial}{\partial p}\right)_T. \tag{28.24}$$

Hence, applying this to the quantity $\Delta S = S_\mathrm{v} - S_\mathrm{L} = L/T$ where the subscripts v and L refer to vapour and liquid respectively, we have that

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}T}\left(\frac{L}{T}\right) &= \left(\frac{\partial(\Delta S)}{\partial T}\right)_p + \frac{\mathrm{d}p}{\mathrm{d}T}\left(\frac{\partial(\Delta S)}{\partial p}\right)_T \\ &= \frac{C_{pv} - C_{p\mathrm{L}}}{T} + \left[\left(\frac{\partial S_\mathrm{v}}{\partial p}\right)_T - \left(\frac{\partial S_\mathrm{L}}{\partial p}\right)_T\right]\frac{\mathrm{d}p}{\mathrm{d}T},\end{aligned} \tag{28.25}$$

so that

$$\frac{\mathrm{d}}{\mathrm{d}T}\left(\frac{L}{T}\right) = \frac{C_{pv} - C_{p\mathrm{L}}}{T} - \left[\frac{\partial}{\partial T}(V_\mathrm{v} - V_\mathrm{L})\right]\frac{\mathrm{d}p}{\mathrm{d}T}. \tag{28.26}$$

Using $V_\mathrm{v} \gg V_\mathrm{L}$ and $pV_\mathrm{v} = RT$, we have that

$$\frac{\mathrm{d}}{\mathrm{d}T}\left(\frac{L}{T}\right) = \frac{C_{pv} - C_{p\mathrm{L}}}{T} - \frac{R}{p} \times \frac{Lp}{RT^2}, \tag{28.27}$$

and expanding

$$\frac{\mathrm{d}}{\mathrm{d}T}\left(\frac{L}{T}\right) = \frac{1}{T}\frac{\mathrm{d}L}{\mathrm{d}T} - \frac{L}{T^2}, \tag{28.28}$$

yields

$$\mathrm{d}L = (C_{pv} - C_{p\mathrm{L}})\,\mathrm{d}T, \tag{28.29}$$

so that

$$L = L_0 + (C_{pv} - C_{p\mathrm{L}})T. \tag{28.30}$$

Thus the latent heat contains a linear temperature dependence and this is shown by the solid line in Fig. 28.4. The negative slope is due to the fact that $C_{p\mathrm{L}} > C_{pv}$. Substituting this value of L into eqn 28.19 yields the equation of the phase boundary:

$$p = p_0 \exp\left(-\frac{L_0}{RT} + \frac{(C_{pv} - C_{p\mathrm{L}})\ln T}{R}\right). \tag{28.31}$$

We can also use the Clausius–Clapeyron equation to derive the phase boundary of the liquid–solid coexistence line, as shown in the following example.

Example 28.4

Find the equation in the p–T plane for the phase boundary between the liquid and solid phases of a substance.
Solution:
The Clausius–Clapeyron equation (eqn 28.19) can be rearranged to give

$$\mathrm{d}p = \frac{L\mathrm{d}T}{T\Delta V}, \tag{28.32}$$

and neglecting the temperature dependence of L and ΔV, we find that this integrates to

$$p = p_0 + \frac{L}{\Delta V}\ln\left(\frac{T}{T_0}\right), \tag{28.33}$$

where T_0 and p_0 are constants such that $(T, p) = (T_0, p_0)$ is a point on the phase boundary. The volume change ΔV on melting is relatively small, so that the gradient of the phase boundary in the p–T plane is very steep.

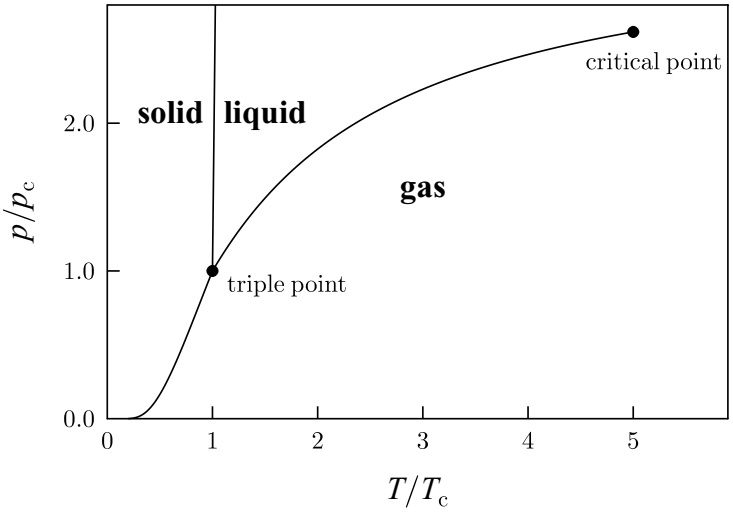


Fig. 28.6 A schematic phase diagram of a (hypothetical) pure substance.

A phase diagram of a hypothetical pure substance is shown in Fig. 28.6 and shows the solid, liquid and gaseous phases coexisting with the phase boundaries calculated from the Clausius–Clapeyron equation. The three

phases coexist at the **triple point**. The solid-liquid phase boundary is very steep, reflecting the large change in entropy in going from liquid to solid and the very small change in volume. This phase boundary does not terminate, but continues indefinitely. By way of contrast, the phase boundary between liquid and gas terminates at the **critical point**, as we have seen in Section 26.1. (We will have more to say about this observation in Section 28.7.) Note also that, at temperatures close to the triple point, the latent heat of **sublimation** (changing from solid to gas) is equal to the sum of the latent heat of melting (solid→liquid)[7] and the latent heat of vapourisation (liquid→gas).[8]

[7]The latent heat of melting is sometimes known as the latent heat of fusion.

[8]This fact will be used in Exercise 28.5.

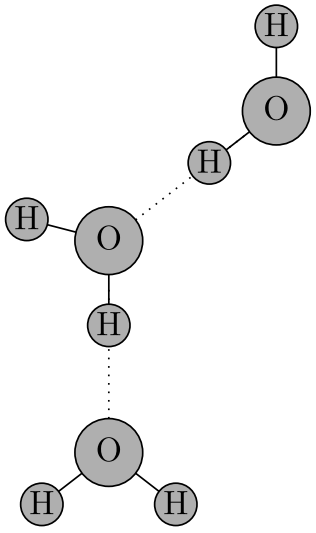


Fig. 28.8 Schematic diagram of hydrogen bonding in water.

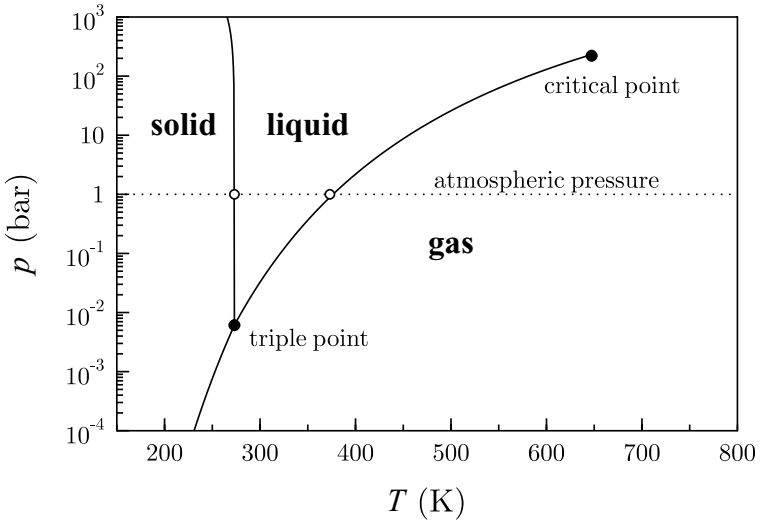


Fig. 28.7 The phase diagram of H_2O showing the solid (ice), liquid (water) and gaseous (steam) phases. The horizontal dashed line corresponds to atmospheric pressure, and the normally experienced freezing and boiling points of water are indicated by the open circles.

The gradient of the liquid–solid coexistence line is normally positive because most substances expand when they melt. A notable counterexample is water, which slightly shrinks when it melts. Hence, the gradient of the ice-water coexistence line is negative (see Fig. 28.7; because the solid-liquid line is so steep, it is not easy to see that the slope is negative). This effect occurs because of the **hydrogen bonding**[9]in water (see Fig. 28.8) which results in a rather open structure of the ice crystal lattice. This collapses on melting, resulting in a slightly denser liquid. This result has many consequences: for example, icebergs float on the ocean and ice cubes float in your gin and tonic. The pressure dependence of the coexistence line means that pressing ice can cause it to melt, an effect which is responsible for the movement of glaciers, which can press against rock, melt near the region of contact with the rock, and slowly creep downhill.

[9]A hydrogen bond is a weak attractive interaction between a hydrogen atom and a strongly electronegative atom such as oxygen or nitrogen. The electron cloud around the hydrogen nucleus is attracted by the electronegative atom and leaves the hydrogen with a partial positive charge, and because of its size this results in a large charge density. Hydrogen bonding is responsible for the linking of the base pairs in DNA and the structure of many proteins. It is also responsible for the high boiling point of water (which given its low molecular mass would be expected to boil at much lower temperatures than it does).

28.4 Stability & metastability

We have seen in Section 28.2 that the phase with the lowest chemical potential μ is the most stable. Let us see how the phase transition varies as a function of pressure. Since μ is the Gibbs function per particle, eqn 16.24 implies that

$$\left(\frac{\partial \mu}{\partial p}\right)_T = v, \tag{28.34}$$

where v is the volume per particle. Since $v > 0$, the gradient of the chemical potential with pressure must always be positive. The behaviour of μ as a function of pressure as one crosses the phase transition between the liquid and gas phases is shown in Fig. 28.9. This figure shows that the phase which is stable at the highest pressure must therefore have the smallest volume. This of course makes sense since, when you apply large pressure, you expect the smallest space-occupying phase to be the most stable.

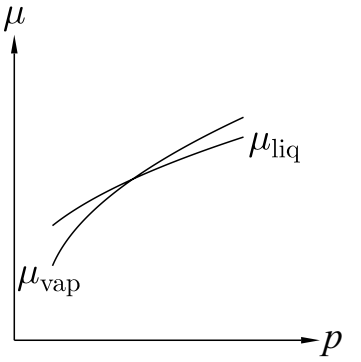


Fig. 28.9 The chemical potential as a function of pressure.

We can also think about μ as a function of temperature. Equation 16.23 implies that

$$\left(\frac{\partial \mu}{\partial T}\right)_p = -s, \tag{28.35}$$

where s is the entropy per particle. Since $\mu > 0$, the gradient of μ as a function of temperature must always be negative. The behaviour of μ as a function of temperature as you cross the phase transition between the liquid and gas phases is shown in Fig. 28.10. This figure shows that the phase which is stable at the highest temperature must therefore have the highest entropy. This makes sense because $G = H - TS$ and so at higher temperature, you minimize G by maximizing S.

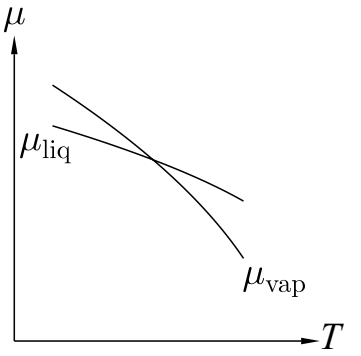


Fig. 28.10 The chemical potential as a function of temperature.

This also shows that as you warm a substance through its boiling point, it is possible to momentarily continue on the curve corresponding to μ_{liq} and to form **superheated liquid** which is a metastable state. Although for temperatures above the boiling point it is the gaseous state which is thermodynamically stable (i.e. has the lowest Gibbs function), there may be reasons why this state cannot be formed immediately and the liquid state persists. Similarly, if you cool a gas below the boiling point, it is possible to momentarily continue on the curve corresponding to μ_{gas} and to form **supercooled vapour**, which is a metastable state. Again, this is not the thermodynamically stable state of the system but there may be reasons why the liquid state cannot nucleate immediately and the gaseous state persists.

Let us now try and fathom the reason why the thermodynamically most stable state sometimes doesn't form. Consider a liquid with pressure p_{liq} in equilibrium with a vapour at pressure p. The chemical potentials of the liquid and vapour must be equal. Now imagine that the liquid pressure slightly increases to $p_{\text{liq}} + \mathrm{d}p_{\text{liq}}$. If the vapour is still in equilibrium with the liquid, then its pressure must increase to $p + \mathrm{d}p$ and we must have that

$$\left(\frac{\partial \mu_{\text{liq}}}{\partial p_{\text{liq}}}\right)_T \mathrm{d}p_{\text{liq}} = \left(\frac{\partial \mu_{\text{vap}}}{\partial p}\right)_T \mathrm{d}p, \tag{28.36}$$

so that the chemical potentials of the liquid and vapour are still equal. Using eqn 28.34, this implies that

$$v_{\text{liq}}\mathrm{d}p_{\text{liq}} = v_{\text{vap}}\mathrm{d}p, \tag{28.37}$$

where v_{liq} is the volume per particle occupied by the liquid and v_{liq} is the volume per particle occupied by the gas. Hence multiplying this by N_{A} and using $pV = RT$ for one mole of the gas, we find that

$$V_{\text{liq}}\mathrm{d}p_{\text{liq}} = \frac{RT\mathrm{d}p}{p}, \tag{28.38}$$

where V_{liq} is the molar volume of the liquid. We can use this to find the dependence of the vapour pressure[10] on the pressure in the liquid at constant temperature. Integrating eqn 28.38 leads to

$$p = p_0 \exp\left(\frac{V_{\text{liq}}\Delta p_{\text{liq}}}{RT}\right), \tag{28.39}$$

where Δp_{liq} is the extra pressure applied to the liquid, p_0 is the vapour pressure of the gas with no excess pressure applied to the liquid and p is the vapour pressure of the gas with excess pressure Δp_{liq} in the liquid.

This result can be used to derive the vapour pressure of a droplet of liquid. Recall from eqn 17.18 that the excess pressure in a droplet of liquid of radius r can be obtained as

$$\Delta p_{\text{liq}} = \frac{2\gamma}{r}, \tag{28.40}$$

where γ is the surface tension. Hence we find that

$$\boxed{p = p_0 \exp\left(\frac{2\gamma V_{\text{liq}}}{rRT}\right),} \tag{28.41}$$

which is known as **Kelvin's formula**. This formula shows that small droplets have a very high vapour pressure, and this gives some understanding about why the vapour sometimes doesn't condense when you cool it through the boiling temperature. Small droplets initially begin to nucleate, but have a very high vapour pressure and therefore instead of growing can evaporate. This stabilises the vapour, even though it is the thermodynamically stable phase. The thermodynamic driving force to condense is overcome by the tendency to evaporate. This effect occurs very often in the atmosphere which contains water vapour which has risen to an altitude where it is sufficiently cold to condense into water droplets, but the droplets cannot form owing to this tendency to evaporate. Clouds do form through the nucleation of droplets on minute dust particles which have sufficient surface area for the liquid to condense and then grow above the critical size.

A similar effect occurs for superheated liquids. The pressure of liquid near a vapour-filled cavity of radius r is less than that in the bulk liquid according to

$$\Delta p_{\text{liq}} = -\frac{2\gamma}{r}, \tag{28.42}$$

[10] The **vapour pressure** of a liquid (or a solid) is the pressure of vapour in equilibrium with the liquid (or the solid).

and hence the vapour pressure inside the cavity follows:

$$p = p_0 \exp\left(-\frac{2\gamma V_{\text{liq}}}{rRT}\right). \tag{28.43}$$

Thus the vapour pressure inside a cavity is lower than one might expect. As you boil a liquid, any bubble of vapour which does form tends to collapse. This means the liquid can become superheated and kinetically stable above its boiling point, even though the vapour is the true thermodynamic ground state. The only bubbles which then do survive are very large ones, and this causes the violent bumping which can be observed in boiling liquids. This can be avoided by boiling liquids with small pieces of glass or ceramic, so that there are plenty of nucleation centres for small bubbles to form.

Example 28.5

A **bubble chamber** is used in particle physics to detect electrically charged subatomic particles. It consists of a container filled with a superheated transparent liquid such as liquid hydrogen, at a temperature just below its boiling point. The motion of the charged particle is sufficient to nucleate a string of bubbles of vapour which display the track of the particle. A magnetic field can be applied to the chamber so that the shape of the curved tracks of the particle can be used to infer its charge to mass ratio. Its invention in 1952 earned Donald Glaser (1926–) the 1960 Nobel Prize for Physics.

Example 28.6

Calculate the Gibbs function for a droplet of liquid of radius r (and hence surface area $A = 4\pi r^2$) in equilibrium with vapour. Assume the temperature is such that the liquid is the thermodynamically stable phase.

Solution:

Writing the number of particles (of mass m) in the liquid and vapour as N_{liq} and N_{vap} respectively, the change in Gibbs function is

$$\mathrm{d}G = \mu_{\text{liq}}\,\mathrm{d}N_{\text{liq}} + \mu_{\text{vap}}\,\mathrm{d}N_{\text{vap}} + \gamma\,\mathrm{d}A, \tag{28.44}$$

where γ is the surface tension. Since particles must be conserved, $\mathrm{d}N_{\text{vap}} = -\mathrm{d}N_{\text{liq}}$. Differentiating $A = 4\pi r^2$ yields $\mathrm{d}A = 8\pi r\,\mathrm{d}r$, and writing $\Delta\mu = \mu_{\text{vap}} - \mu_{\text{liq}}$ (which will be positive since the liquid is the thermodynamically stable phase) we have

$$\mathrm{d}G = \left(8\pi\gamma r - \frac{4\pi r^2 \Delta\mu \rho_{\text{liq}}}{m}\right)\mathrm{d}r, \tag{28.45}$$

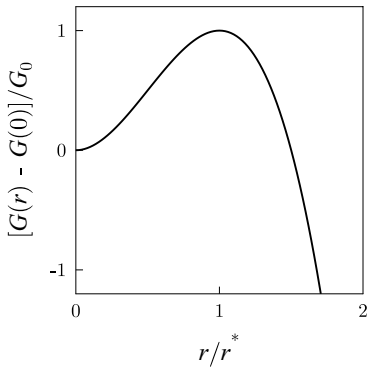


Fig. 28.11 The Gibbs function of the droplet as a function of radius, plotted in units of $G_0 = 16\pi\gamma^3 m^2/(3(\rho_{\rm liq}\Delta\mu)^2)$.

where $\rho_{\rm liq}$ is the density of the liquid. This can be integrated to yield

$$G(r) = G(0) + 4\pi\gamma r^2 - \frac{4\pi\Delta\mu\rho_{\rm liq}}{3m}r^3, \tag{28.46}$$

and hence equilibrium is established when $\mathrm{d}G/\mathrm{d}r = 0$, and this occurs at the critical radius r^* given by

$$r^* = \frac{2\gamma m}{\rho_{\rm liq}\Delta\mu}. \tag{28.47}$$

This function is sketched in Fig. 28.11 and shows that r^* is indeed a stationary point, but is a maximum in G, not a minimum! Thus $r = r^*$ is a point of unstable equilibrium. If $r < r^*$, the system can minimize G by shrinking r to zero, i.e. the droplet evaporates. If $r > r^*$, the system can minimize G by the droplet growing to infinite size.

This effect occurs as water condenses in a cloud. The large droplets keep the partial pressure of the water vapour low. The smaller droplets therefore evaporate and the water can transfer from the smaller to the larger droplets.

28.5 The Gibbs phase rule

In this section, we want to find out how much freedom a system has to change its internal parameters while keeping the different substances in various combinations of phases in equilibrium with each other. We want to include the possibility of having mixtures of different substances, and we will call the different substances *components*. A component is a chemically independent constituent of the system. To keep track of the number of molecules in these different components, we introduce the **mole fraction** x_i which is defined to be the ratio of the number of moles, n_i, of the ith substance, divided by the total number of moles n, so that

$$x_i = \frac{n_i}{n}. \tag{28.48}$$

By definition, we have that

$$\sum x_i = 1. \tag{28.49}$$

Each of the components can be in different *phases* (where here we mean phases such as 'solid', 'liquid' and 'gas', but we might also wish to include other possibilities, such as 'ferromagnetic' and 'paramagnetic' phases, or 'superconducting' and 'non-superconducting' phases). We denote by the symbol F the number of degrees of freedom the system has while keeping the different phases in equilibrium, and it is this quantity we now want to calculate, following a method introduced by Gibbs.

Consider a multicomponent system, containing C components. Each component can be in any one of P different phases. The system is

characterized by the intensive variables, the pressure p, the temperature T and the mole fractions of $C-1$ of the components (we don't need all C of them, since $\sum_{i=1}^{C} x_i = 1$) for each of the P phases, so that is

$$2 + P(C-1) \tag{28.50}$$

variables. If the phases of each component are in equilibrium with one another, then we must have, as i runs from 1 to C,

$$\mu_i(\text{phase } 1) = \mu_i(\text{phase } 2) = \cdots = \mu_i(\text{phase } P), \tag{28.51}$$

which gives us $P-1$ equations to solve for each component, and hence $C(P-1)$ equations to solve for each of the C components.

The number of degrees of freedom F the system has is given by the difference between the number of variables and the number of constraining equations to solve. Hence $F = [P(C-1)+2] - C(P-1)$, and thus

$$\boxed{F = C - P + 2} \tag{28.52}$$

which is known as the **Gibbs phase rule**.

Example 28.7

For a single–component system, $C = 1$ and hence $F = 3 - P$. Thus:

- If there is one phase, $F = 2$, and the whole p–T plane is accessible.
- If there are two phases, $F = 1$, and these two phases can only coexist at a common *line* of coexistence in the p–T plane.
- If there are three phases, $F = 0$, and these three phases can only coexist at a common *point* of coexistence in the p–T plane (the triple point).

For a two–component system, $C = 2$ and hence $F = 4 - P$. If we fix the pressure, then the number of remaining degrees of freedom $F' = F - 1 = 3 - P$. Having fixed the pressure, we have two variables which are temperature T and the mole fraction x_1 of the first component (the mole fraction of the second component being given by $1 - x_1$). Thus:

- If there is one phase, $F = 2$, and the whole x_1–T plane is accessible.
- If there are two phases, $F = 1$, and these two phases can only coexist at a common *line* of coexistence in the x_1–T plane.
- If there are three phases, $F = 0$, and these three phases can only coexist at a common *point* of coexistence in the x_1–T plane.

The Gibbs phase rule is of great use in interpreting complex phase diagrams of mixtures of substances.

28.6 Colligative properties

When a liquid of a particular material (we will call it A) has another species, B, dissolved in it, the chemical potential of A is decreased. The result of this is that the boiling point of the liquid A is elevated and the freezing point of the liquid is depressed compared to that of the pure liquid. These effects are known as **colligative properties**.[11] The magnitude of the effect can be worked out from the reduction of the chemical potential.

[11] The word colligative means a collection of things fastened together.

Example 28.8

Find the chemical potential of a solvent A with a solute B dissolved in it. The mole fraction of the solvent is x_{A}.

The liquid which is the main component is known as the **solvent**, while the material which is dissolved in it is known as the **solute**.

Solution:

Recall from eqn 22.65 that the chemical potential of a gas (let us call it a gas of molecules of A) with pressure p^*_{A} is given by

$$\mu^{(\mathrm{g})*}_{\mathrm{A}} = \mu^{\ominus}_{\mathrm{A}} + RT \ln \frac{p^*_{\mathrm{A}}}{p^{\ominus}}, \tag{28.53}$$

where the superscript (g) indicates gas and the superscript * indicates that we are dealing with a pure substance. If this is in equilibrium with the liquid form of A, then we also have

$$\mu^{(\ell)*}_{\mathrm{A}} = \mu^{\ominus}_{\mathrm{A}} + RT \ln \frac{p^*_{\mathrm{A}}}{p^{\ominus}}, \tag{28.54}$$

where the superscript (ℓ) indicates liquid. Now imagine that we mix some B molecules into the liquid. The mole fraction of A, x_{A}, is now less than one. The chemical potential of A in the liquid is now still equal to the chemical potential of A in the gas, but the gas has a different vapour pressure p_{A} (no asterisk because we are no longer dealing with pure substances). Thus

$$\mu^{(\ell)}_{\mathrm{A}} = \mu^{(\mathrm{g})}_{\mathrm{A}} = \mu^{\ominus}_{\mathrm{A}} + RT \ln \frac{p_{\mathrm{A}}}{p^{\ominus}}. \tag{28.55}$$

Equations 28.54 and 28.55 give that

$$\mu^{(\mathrm{g})}_{\mathrm{A}} = \mu^{(\ell)*}_{\mathrm{A}} + RT \ln \frac{p_{\mathrm{A}}}{p^*_{\mathrm{A}}}. \tag{28.56}$$

The vapour pressure of A in the mixed system can be estimated using **Raoult's law**, which states that $p_{\mathrm{A}} = x_{\mathrm{A}} p^*_{\mathrm{A}}$ (i.e. that the vapour pressure of A is proportional to its mole fraction). Hence eqn 28.56 becomes

$$\mu^{(\ell)}_{\mathrm{A}} = \mu^{(\ell)*}_{\mathrm{A}} + RT \ln x_{\mathrm{A}}. \tag{28.57}$$

Since $x_{\mathrm{A}} < 1$, we find that $\mu^{(\ell)}_{\mathrm{A}} < \mu^{(\ell)*}_{\mathrm{A}}$ and so the chemical potential is indeed depressed compared to the pure case.

We can now derive formulae to describe the colligative properties. Equation 28.57 can be rewritten as

$$\ln x_{\mathrm{A}} = \frac{\Delta G_{\mathrm{vap}}}{RT}, \tag{28.58}$$

where $\Delta G_{\mathrm{vap}} = \mu_{\mathrm{A}}^{(g)*} - \mu_{\mathrm{A}}^{(\ell)*}$. When $x_{\mathrm{A}} = 1$, then equilibrium between vapour and liquid occurs at a temperature T^* given by (using eqn 28.58 with $x_{\mathrm{A}} = 1$)

$$\frac{\Delta G_{\mathrm{vap}}(T^*)}{RT^*} = 0, \tag{28.59}$$

which implies that (recall that $G = H - TS$)

$$\Delta H_{\mathrm{vap}}(T^*) - T^* \Delta S_{\mathrm{vap}}(T^*) = 0. \tag{28.60}$$

When $x_{\mathrm{B}} = 1 - x_{\mathrm{A}}$ is very small, then we have that

$$\ln(1 - x_{\mathrm{B}}) \approx -x_{\mathrm{B}}, \tag{28.61}$$

and hence eqn 28.58 implies that

$$-x_{\mathrm{B}} = \frac{\Delta G_{\mathrm{vap}}}{RT} = \frac{1}{R}\left[\Delta H_{\mathrm{vap}}(T) - T\Delta S_{\mathrm{vap}}(T)\right], \tag{28.62}$$

and assuming that ΔH_{vap} and ΔS_{vap} are only weakly temperature-dependent, this yields

$$-x_{\mathrm{B}} = \frac{\Delta H_{\mathrm{vap}}}{R}\left(\frac{1}{T^*} - \frac{1}{T}\right) \approx \frac{\Delta H_{\mathrm{vap}}\delta}{RT^{*2}}(T - T^*). \tag{28.63}$$

Hence $T - T^*$, the elevation in boiling point, is given approximately by

$$T - T^* \approx \frac{RT^{*2}x_{\mathrm{B}}}{\Delta H_{\mathrm{vap}}}. \tag{28.64}$$

It is often written $T - T^* = K_{\mathrm{b}}x_{\mathrm{B}}$, where $K_{\mathrm{b}} \approx RT^{*2}/\Delta H_{\mathrm{vap}}$ is known as the **ebullioscopic constant**. For water, $K_{\mathrm{b}} = 0.51\,\mathrm{K\,mol^{-1}\,kg^{-1}}$. There is a similar effect on the depression of the freezing point. One can show similarly that the freezing point is depressed by an amount $T^* - T = K_{\mathrm{f}}x_{\mathrm{B}}$ where $K_{\mathrm{f}} \approx RT^{*2}/\Delta H_{\mathrm{fus}}$ is the **cryoscopic constant**. The salt water in the oceans freezes at a lower temperature than fresh water. The effect is also relevant for salt being put on pavements (sidewalks) in winter to stop them becoming icy.

Adding a small quantity of solute to a solvent increases the entropy of the solvent because the solute atoms are randomly located in the solvent. This means that there is a weaker tendency to form a gas (which would increase the solvent's entropy) because the entropy of the solvent has been increased anyway. This results in an elevation of the boiling point. Similarly, this additional entropy opposes the tendency to freeze and the freezing point is depressed.

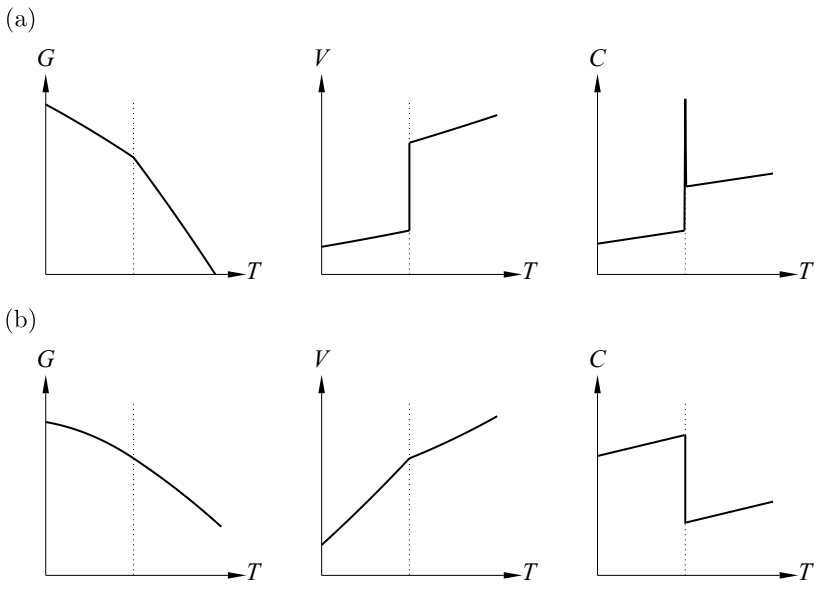


Fig. 28.12 Ehrenfest's classification of phase transitions. (a) First-order phase transition. (b) Second-order phase transition. The critical temperature T_c is marked by a vertical dotted line in each case.

28.7 Classification of phase transitions

Paul Ehrenfest (1880–1933) proposed a classification of phase transitions which goes as follows: the **order** of a phase transition is the order of the lowest differential of G (or μ) which shows a discontinuity at T_c. Thus **first-order phase transitions** involve a latent heat because the entropy (a first differential of G) shows a discontinuity. The volume is also a first differential of G and this also shows a discontinuous jump. The heat capacity is a second differential of G and thus it shows a sharp spike, as does the compressibility. This is illustrated in Fig. 28.12(a). Examples of first-order phase transitions include the solid–liquid transition, the solid–vapour transition, and the liquid–vapour transition.

By Ehrenfest's classification, a **second-order phase transition** has no latent heat because the entropy does not show a discontinuity (and neither does the volume – both are first differentials of G), but quantities like the heat capacity and compressibility (second differentials of G) do. This is illustrated in Fig. 28.12(b). Examples of second-order phase transitions include the superconducting transition, or the order–disorder transition in β-brass.

However, a big problem with the approach we have been using so far in studying phase transitions is that one key approximation made in thermodynamics, namely that the number of particles is so large that average properties such as pressure and density are well defined, breaks down at a phase transition. Fluctuations build up near a phase transition and so the behaviour of the system does not follow the expectations of our analysis very close to the phase transition temperature. This critical region is characterized by fluctuations, at all length scales. For example, when a saucepan of water is heated, the water warms quite quietly and

unobtrusively until near the boiling point when it makes a great deal of noise and bubbles violently.[12] We have already analysed the behaviour of the formation of bubbles in Section 28.4. Therefore, it has been found that Ehrenfest's approach is rather too simple. We will have more to say concerning fluctuations in Chapters 33 and 34.

A more modern approach to classifying phase transitions simply distinguishes between those which show a latent heat, for which Ehrenfest's term "first-order phase transition" is retained, and those which do not, which are called a **continuous phase transition** (and include Ehrenfest's phase transitions of second order, third order, fourth order, etc, all lumped together).

[12] A visual demonstration of this is found in the phenomenon known as **critical opalescence** which is the blurring and clouding of images seen through a volume of gas near its critical point. This occurs because density fluctuations are strong near the critical point and give rise to large variations in refractive index.

Example 28.9

- The liquid–gas phase transition is a first-order transition, except at the critical point where the phase transition involves no latent heat and is a continuous phase transition.
- A ferromagnet[13] such as iron loses its ferromagnetism when heated to the Curie temperature, T_C (a particular example of a critical temperature). This phase transition is a continuous phase transition, since there is no latent heat. The magnetization is a first differential of the Gibbs function and does not change discontinuously at T_C. The specific heat C_B, at constant magnetic field B, has a finite peak at T_C.

[13] A ferromagnet is a material containing magnetic moments which are all aligned in parallel below a transition temperature called the Curie temperature. Above this temperature, the magnetic moments become randomly aligned. This state is known as the paramagnetic state.

A further classification of phase transitions involves the notion of **symmetry breaking**. Figure 28.13 shows atoms in a liquid and in a solid. As a liquid cools there is a very slight contraction of the system but it retains a very high degree of symmetry. However, below the melting temperature, the liquid becomes a solid and that symmetry is broken. This may at first sight seem surprising because the picture of the solid 'looks' more symmetrical than that of the liquid. The atoms in the solid are all symmetrically lined up while in the liquid they are all over the place. The crucial observation is that any point in a liquid is, on average, exactly the same as any other. If you average the system over time, each position is visited by atoms as often as any other. There are no unique directions or axes along which atoms line up. In short, the system possesses complete translational and rotational symmetry. In the solid, however, this high degree of symmetry is nearly all lost. The solid drawn in Fig. 28.13 still possesses some residual symmetry: rather than being invariant under arbitrary rotations, it is invariant under four-fold rotations ($\pi/2$, π, $3\pi/2$, 2π); rather than being invariant under arbitrary translations, it is now invariant under a translation of

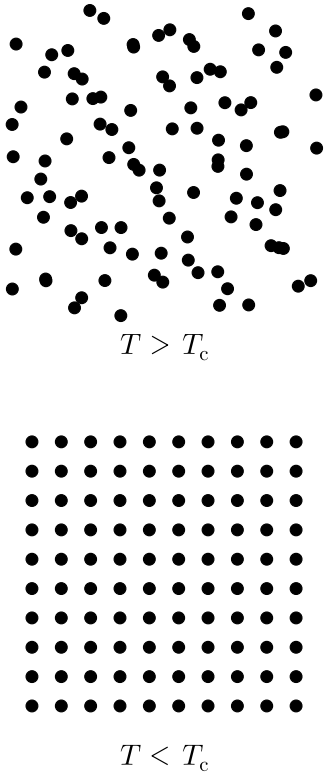


Fig. 28.13 The liquid–solid phase transition. *Top:* The high temperature state (statistically averaged) has complete translational and rotational symmetry. *Bottom:* These symmetries are broken as the system becomes a solid below the critical temperature T_c.

[14]In other words when $k_B T$ was lower than this energy, corresponding to a temperature $T \sim 10^{15}$ K.

an integer combination of lattice basis vectors. Therefore not all symmetry has been lost but the high symmetry of the liquid state has been, to use the technical term, 'broken'. It is impossible to change symmetry *gradually*. Either a particular symmetry is present or it is not. Hence, phase transitions are sharp and there is a clear delineation between the ordered and disordered states.

Not all phase transitions involve a change of symmetry. Consider the liquid–gas coexistence line again (see Fig. 28.7). The boundary line between the liquid and gas regions is terminated by a critical point. Hence it is possible to 'cheat' the sharp phase transition by taking a path through the phase diagram which avoids a discontinuous change. For temperatures above the critical temperature (647 K for water) the gaseous and liquid states are distinguished only by their density. The transition between a gas and a liquid involves no change of symmetry and therefore it is possible to avoid it by working round the critical end point. In contrast, the solid–liquid transition involves a change of symmetry and consequently there is no critical point for the melting curve.

Symmetry-breaking phase transitions include those between the ferromagnetic and paramagnetic states (in which the low–temperature state does not possess the rotational symmetries of the high–temperature state) and those between the superconducting and normal metal states of certain materials (in which the low–temperature state does not possess the same symmetry in the phase of the wavefunction as the high temperature state).

The concept of broken symmetry is very wide–ranging and is used to explain how the electromagnetic and weak forces originated. In the early Universe, when the temperature was very high, it is believed that the electromagnetic and weak forces were part of the same, unified, electroweak force. When the temperature cooled[14] to below about 10^{11} eV a symmetry was broken and a phase transition occured, via what is known as the **Higgs mechanism**, and the W and Z bosons (mediating the weak force) acquired mass while the photon (mediating the electromagnetic force) remained massless. It is suggested that, at even earlier times, when the temperature of the Universe was around 10^{21} eV, the electroweak and strong forces were unified, and as the Universe expanded and its temperature lowered, another symmetry-breaking transition caused them to appear as different forces.

Chapter summary

- The latent heat is related to the change in entropy at a first–order phase transition.
- The Clausius–Clapeyron equation states that

$$\frac{\mathrm{d}p}{\mathrm{d}T} = \frac{L}{T(V_2 - V_1)},$$

  and this can be used to determine the shape of the phase boundary.
- The Kelvin formula states that the pressure in a droplet is given by

$$p = p_0 \exp\left(\frac{2\gamma V_{\mathrm{liq}}}{rRT}\right).$$

- The Gibbs phase rule states that $F = C - P + 2$.
- Dissolving a solute in a solvent results in the elevation of the solvent's boiling point and a depression of its freezing point.
- A first–order phase transition involves a latent heat while a continuous phase transition does not.
- Certain phase transitions involve the breaking of symmetry.

Further reading

More information on phase transitions may be found in Binney *et al.* (1992), Yeomans (1992), Le Bellac (2004), Blundell (2001) and Anderson (1984).

Exercises

(28.1) When lead is melted at atmospheric pressure, the melting point is 327.0°C, the density decreases from 1.101×10^4 to 1.065×10^4 kg m^{-3} and the latent heat is 24.5 kJ kg^{-1}. Estimate the melting point of lead at a pressure of 100 atm.

(28.2) Some tea connoisseurs claim that a good cup of tea cannot be brewed with water at a temperature less than 97°C. Assuming this to be the case, is it possible for an astronomer, working on the summit of Mauna Kea in Hawaii (elevation 4194 m, though you don't need to know this to solve the problem) where the air pressure is 615 mbar, to make a good cup of tea without the aid of a pressure vessel?

(28.3) The gradient of the melting line of water on a p–T diagram close to 0° C is -1.4×10^7 Pa K^{-1}. At 0° C, the specific volume of water is 1.00×10^{-3} m^3 kg^{-1} and of ice is 1.09×10^{-3} m^3 kg^{-1}. Using this information, deduce the latent heat of fusion of ice.
In winter, a lake of water is covered initially by a uniform layer of ice of thickness 1 cm. The air temperature at the surface of the ice is −0.5°C. Es-

timate the rate at which the layer of ice begins to thicken, assuming that the temperature of the water just below the ice is 0°C. You can also assume steady state conditions and ignore convection.
The temperature of the water at the bottom of the lake, depth 1 m, is maintained at 2°C. Find the thickness of ice which will eventually be formed.
[The thermal conductivity of ice is $2.3\,\mathrm{W\,m^{-1}\,K^{-1}}$ and of water is $0.56\,\mathrm{W\,m^{-1}\,K^{-1}}$.]

(28.4) (a) Show that the temperature dependence of the latent heat of vaporization L is given by the following expression:

$$\frac{\mathrm{d}}{\mathrm{d}T}\left(\frac{L}{T}\right) = \frac{C_{pv}-C_{pL}}{T} + \left[\left(\frac{\partial S_v}{\partial p}\right)_T - \left(\frac{\partial S_L}{\partial p}\right)_T\right]\frac{\mathrm{d}p}{\mathrm{d}T}. \quad (28.65)$$

In this equation, S_v and S_L are the entropies of the vapour and liquid and C_{pv} and C_{pL} are the heat capacities of the vapour and liquid. Hence show that $L = L_0 + L_1 T$ where L_0 and L_1 are constants.
(b) Show further that when the saturated vapour of an incompressible liquid is expanded adiabatically, some liquid condenses out if

$$C_{pL} + T\frac{\mathrm{d}}{\mathrm{d}T}\left(\frac{L}{T}\right) < 0$$

where C_{pL} is the heat capacity of the liquid (which is assumed constant) and L is the latent heat of vaporisation.
(Hint: consider the gradient of the phase boundary in the p–T plane and the corresponding curve for adiabatic expansion.)

(28.5) The equilibrium vapour pressure p of water as a function of temperature is given in the following table:

T (°C)	p (Pa)
0	611
10	1228
20	2339
30	4246
40	7384
50	12349

Deduce a value for the latent heat of evaporation L_v of water. State clearly any simplifying assumptions that you make.
Estimate the pressure at which ice and water are in equilibrium at −2°C given that ice cubes float with 4/5 of their volume submerged in water at the triple point (0.01°C, 612 Pa).
[Latent heat of sublimation of ice at the triple point, $L_s = 2776 \times 10^3\,\mathrm{J\,kg^{-1}}$.]

(28.6) It is sometimes stated that the weight of a skater pressing down on their thin skates is enough to melt ice, so that the skater can glide around on a thin film of liquid water. Assuming an ice rink at −5°C, do some estimates and show that this mechanism won't work. [In fact, frictional heating of ice is much more important, see S.C. Colbeck, *Am. J. Phys.* **63**, 888 (1995) and S. C. Colbeck, L. Najarian, and H. B. Smith *Am. J. Phys.* **65**, 488 (1997).]

Bose–Einstein and Fermi–Dirac distributions

29

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In this chapter, we are going to consider the way in which *quantum mechanics* changes the statistical properties of gases. The crucial ingredient is the concept of *identical particles*. The results of quantum mechanics show that there are two types of identical particle: **bosons** and **fermions**. Bosons can share quantum states, while fermions cannot share quantum states. Another way of stating this is to say that bosons are not subject to the **Pauli exclusion principle**, while fermions are. This difference in ability to share quantum states (arising from what we shall call *exchange symmetry*) has a profound effect on the statistical distribution of these particles over the energy states of the system. This distribution over energy states is called the **statistics** of these particles, and we will demonstrate the effect of exchange symmetry on statistics. However, it can also be shown that another difference between bosons and fermions is the type of spin angular momentum that they may possess. This is enshrined in the **spin-statistics theorem**, which we will not prove but which states that bosons have integer spin while fermions have half-integer spin.

Example 29.1

- Examples of bosons include: photons (spin 1), ^{4}He atoms (spin 0).
- Examples of fermions include: electrons (spin $\frac{1}{2}$), neutrons (spin $\frac{1}{2}$), protons (spin $\frac{1}{2}$), ^{3}He atoms (spin $\frac{1}{2}$), ^{7}Li nuclei (spin $\frac{3}{2}$).

29.1 Exchange and symmetry

In this section, we will argue why a two-particle wave function can be either symmetric or antisymmetric under exchange of particles. Consider two identical particles, one at position $\boldsymbol{r}_1$ and the other at position $\boldsymbol{r}_2$. The wave function which describes this is $\psi(\boldsymbol{r}_1, \boldsymbol{r}_2)$. We now define an exchange operator $\hat{P}_{12}$ which exchanges particles 1 and 2. Thus

$$\hat{P}_{12}\psi(\boldsymbol{r}_1, \boldsymbol{r}_2) = \psi(\boldsymbol{r}_2, \boldsymbol{r}_1). \tag{29.1}$$